

reference format

Font format: Arial, Front size:11

Date:

Title:

Purpose:

Procedure:

1. Heat the top of the glass vial with the Bunsen burner.
2. Add sterile water and crack it with tweezers.
3. Remove paper and cotton.
4. Add 500 μL LB medium and resuspend the tablet.
5. Drop the supernatant and streak it with four-quadrant streak method.
6. Spread 100 μL supernatant on LB plate with 100 μL 25 mg/mL Chloramphenicol.
7. Inoculate the remaining supernatant in 5 mL LB medium with 25 $\mu\text{g/mL}$ Chloramphenicol.
8. Incubate in 37°C, started from 18:00

Done by Chars, recorded by Chars.

Results:

June

2026/6/1



Date: 20260601

Title: Growth Curve of *Bacillus subtilis* 168 in normoxia and hypoxia conditions.

Purpose: To test the growth curve of *Bacillus subtilis* 168 in different conditions.

Procedure:

1. Culture the *Bacillus subtilis* 168 in culture tube overnight.
2. Check the OD₆₀₀ (2.04 A).
3. Dilute the *Bacillus subtilis* to 1:25 in 40 mL LB.
4. Culture the cells in 37°C with 200 rpm shaker; different conditions were performed.
5. Measure the OD₆₀₀ per hour.

Results:

Hours	Normoxia (A)	Hypoxia (A)
1	0.175	0.192
2	0.658	0.658
3	1.132	1.045
4	1.461	1.346
5	1.502	1.410
5.75	1.476	1.440

Done by Gabriel.

ELN by Sophia Y.

Date: 20260601

Title: Hydroponics medium

Purpose: Prepare ½ MS medium for hydroponics

Procedure:

1. Measure 1.15g of ½ MS medium without sucrose in 1000mL de-ionized water and mix with a stirring rod.

Done by Chloe , Sophie C

2026/6/2



Date: 20260602

Title: Transformation

Purpose: Pick up single column for sequencing

Procedure:

1. Add 5 μL of 10 ng/ μL of Csn:AmilCP and CSN ACCD to 100 μL competent cell
Add 5 μL of 100 ng/ μL of Csn:AmilCP and CSN ACCD to 100 μL competent cell
2. Incubate in ice for 20 min
3. Heatshock of 42°C 45 sec
4. Incubate in ice for 3 min
5. Add 1000 μL recovery medium in each incubate
6. Place in incubator 37°C 48 min
7. Spin down at 5000 g for 5 min
8. Taking out 1000 μL and resuspend the cell
9. Speeding on the LB agar plate with ampicillin
10. Place in incubate 37°C over night (16 hr)

Done by Sarah C , Olivia L , Sophie C

Date: 20260602

Title: DNA purification of promoter-PcpcG2-veg & EEED

Purpose: Isolate and concentrate pure promoter

Procedure:

1. Pipette 100 μL PCR products in each of four 1.5 mL centrifuge tubes.
2. Add 500 μL of GP buffer to each tube and mix well.
3. Place a GP column into a collection tube → add all the mixture from step 1 into the tube.
4. Load no more than 700 μL into the column each time.
5. Centrifuge 1 min + discard flow through.
6. Add 500 μL WN buffer.
7. Centrifuge 1 min + discard flow through.
8. Add 500 μL WS buffer.
9. Centrifuge 1 min + discard flow through.
10. Centrifuge 8 min at 23444 g and discard flow through.
11. Place the column into a new microcentrifuge tube.
12. Add 30 μL of nuclease free water into the center of the membrane without touching the membrane.
13. Incubate for 3 min and centrifuge at full speed for 2 min to elute DNA.
14. Measure the concentration of DNA and label on the tube. Store DNA at -20°C until use.

Done by Chloe.

ELN by Sophia Y.

Date: 20260602

Title: PCR of PcpcG2-veg & EEBD

Purpose: To magnify the number of desired DNA sequence through replication in the PCR machine

Procedure:

1. Prepare the following.

Reagent	1 rxn (50 μ L)	6 rxn (300 μ L)
10x ExTaq Buffer	5 μ L	30 μ L
2.5 μ M dNTP mix	4 μ L	24 μ L
10 μ M forward primer	2 μ L	6 μ L
10 μ M reverse primer	1 μ L	6 μ L
DNA temp (10 ng/ μ L)	1 μ L	6 μ L
PCR grade water	37.75 μ L	226.5 μ L
ExTaq polymerase	0.25 μ L	1.5 μ L

Done by Chloe and Sophie C.

Supervised by Gabriel.

ELN by Sophia Y. Sophie H.

2026/6/3



Date: 20260603

Title: Transformation

Purpose: Pick up single column for sequencing

Procedure:

1. Add 2 μL of 10 ng/ μL of SamyQ: AmilCP, Ho1: pcyA, ccaSm3, ccaR, and Csn:Lea to 30 μL competent cell.
2. Incubate in ice for 20 min.
3. Heatshock 42°C for 45 sec.
4. Incubate in ice for 3 min.
5. Spread on the LB agar plate with Ampicillin.
6. Place in an incubator at 37 deg C overnight (16 hr).

Done by Sophie L, Sophie C, Sophia Y.

Supervised by Chars.

ELN by Sophia Y.

Date: 20260603

Title: Liquid culture

Purpose: Grow and amplify bacterial cells from single colonies

Procedure:

1. Prepare TB medium: 12 mL of TB medium and 12 μL of 50mg/mL Ampicillin.
2. Dispense 3 mL of medium in each tube.
3. Inoculate a colony into each tube.

Done by Sophie L, Sophie C.

Supervised by Chars.

ELN by Sophia Y.

2026/6/4 

Date: 20260604

Title: Miniprep

Purpose: To isolate pTwist-Amp-High-Csn:AmilCP/ pTwist-CSN-ACCD

Procedure:

1. Transfer 1.8mL liquid culture to 2mL microtubes
2. Centrifuge at 16000 g 1min&discard supernatant
3. Repeat step 1&2 steps until transfer all the liquid culture
4. Add 200µL of MX1 buffer into each microtube & resuspend pellet
5. Add 250µL of MX2 buffer and invert 4-6times until solution becomes clear
6. Incubate at room temp for 2 min
7. Add 350µL of MX3 buffer and invert 4-6 times
8. Centrifuge for 5 min at 23444 g
9. Get 700µL of the supernatant and put it in the column
10. Centrifuge for 30sec at 7000 g
11. Repeat steps 9 and 10 until the supernatant is completely used up
12. Add 500µL WN buffer & centrifuge for 30sec at 7000 g
13. Discard the follow through
14. Add 700µL WS buffer & centrifuge for 30sec at 7000 g
15. Discard follow through & centrifuge empty column for 3 min at 10000 g to remove residual ethanol
16. Place the column in an empty 1.5mL microcentrifuge tube and apply 50µL of ddH₂O to the enter of column -> rest 3min & centrifuge for 3min 10000 g

Result:

Biodrop Results	concentration	A260/230	A260/280
pTwist-Amp-CSN:AmilCP1	255ng/µL	1.635	2.143
pTwist-Amp-CSN-ACCD1	239ng/µL	1.691	2.146
pTwist-Amp-CSN:AmilCP2	305ng/µL	1.640	2.148
pTwist-Amp-CSN-ACCD2	278.0ng/µL	1.007	2.014

Done by Chloe, Sophie L, Anton, Noah

Supervised by :Chars

Date: 20260604

Title: Prepare 0.8% Agarose gel for DNA electrophoresis

Purpose: To prepare a 0.8% agarose gel for analytical DNA electrophoresis.

Procedure:

1. Assemble the mold and tray, and prepare the comb
2. Weight 0.16g agarose powder and 20ml 1X TAE solution
3. Add 1X TAE and powder to the conical flask
4. Cover the flask with plastic wrap, use tip to puncture a hole on it
5. Microwave for 30sec , then take it out and agitate it

6. Microwave for 10 sec and agitate it, repeat this step until there is no particle in melted gel
7. Wait until the melted gel stops smoking but is still hot
8. Add 1ml safety dye and agitate it
9. Pour the melted gel into the tray, then assemble the comb

Done by Sophie L, Noah

Supervised by Chars

Recorded by Sophie H.

Date: 20260604

Title: Plasmid Linearization by Enzyme Digestion

Purpose: To prepare a 0.8% agarose gel for analytical DNA electrophoresis.

Procedure:

A: Csn: AmilCP1

B: Csn: ACCD1

C: Csn:AmilCP2

D: Csn:ACCD 2

1. Calculate the reaction formula as follows:

	A	B	C	D
10x cutsmart buffer	2 μ L	2 μ L	2 μ L	2 μ L
EcoRI-HF	0.5 μ L	0.5 μ L	0.5 μ L	0.5 μ L
DNA	2 μ L	1.8 μ L	1.6 μ L	1.8 μ L
ddH₂O	15.5 μ L	15.7 μ L	15.9 μ L	15.7 μ L

2. Prepare the reaction:
 - 2-a: Add water, DNA, and buffer
 - 2-b: Then add enzymes
3. Incubate in PCR machine

Done by Sophie L. , Noah

Supervised by Chars

Recorded by Sophie H.

Date: 20260604

Title: Liquid culture**Purpose: grow and amplify bacterial cells from single colonies****Procedure:**

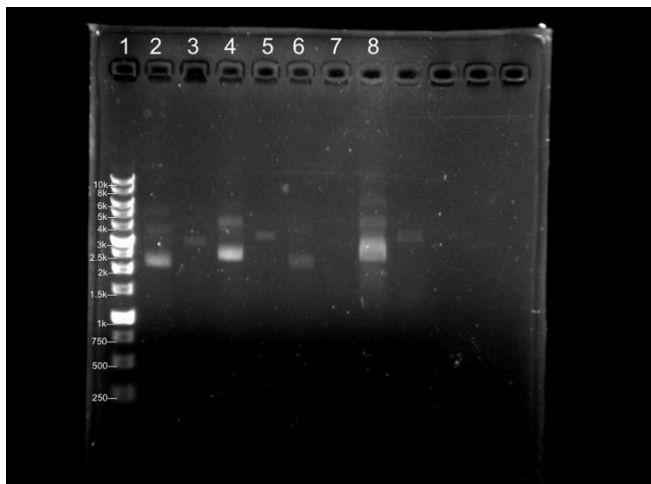
1. Prepare TB medium: 35ml of TB medium and 35 μ L of 50mg/ml Ampicillin
2. Dispense 3ml of medium in each tube
3. Inoculate a colony into each tube
Done by Sophie L, Sophie C, Abby K, Naomi, Joshua, Sophie H.
Supervised by Chars
Recorded by Sophie H.

Date: 20260604**Title: Electrophoresis****Purpose: Verify the gene synthesis produce****Procedure:**

1. Mix 5 μ L DNA(n/ug) with 1 μ L 6x loading dye
2. Loading the sample by the following layout

<u>M</u>	<u>A1</u>	<u>A1</u>	<u>B1</u>	<u>B1</u>	<u>A2</u>	<u>A2</u>	<u>B2</u>	<u>B2</u>	plasmid #clone
5	6	6	6	6	6	6	6	6	volume(μ L)

3. Resolve the DNA by 100v 30 min



- 1: Omics 1kb DNA RTU Ladder 5 μ L
- 2: pTwist-Amp-High-Csn:AmilCP #1 5 μ L (1.2 ug)
3. pTwist-Amp-High-Csn:AmilCP #1 EcoRI 2hr 5 μ L (125 ng) (3051bp)
4. pTwist-Amp-High-Csn:Accd #1 5 μ L (1.4 ug)
5. pTwist-Amp-High-Csn:Accd #1 EcoRI 2hr 5 μ L (125 ng) (3423bp)
6. pTwist-Amp-High-Csn:AmilCP #2 5 μ L (1.5 ug) (3051bp)
7. pTwist-Amp-High-Csn:AmilCP #2 EcoRI 2hr 5 μ L (125 ng) (3051bp)
8. pTwist-Amp-High-Csn:Accd #2 5 μ L (1.4 ug)
9. pTwist-Amp-High-Csn:Accd #2 EcoRI 2hr 5 μ L (125 ng) (3423bp)

Done by Chars

2026/6/5



Date: 20260605

Title: Miniprep

Purpose: isolate plasmid

Procedure:

1. Prepare Glycerol stock, formula as follows:

	pTwist-Amp-S amyQ: AmilCP	pTwist-Amp- Ho1: PoyA	pTwist-Amp -ccaSm3	pTwist-Amp- ccaR	pTwist-Amp- csn: Lea
50% Glycerol Stock	500 µL	500 µL	500 µL	500 µL	500 µL
Bacteria Culture Medium	500 µL	500 µL	500 µL	500 µL	500 µL

Then store in -80 deg C.

2. Transfer 1 mL TB medium to 1.5 mL microtube.
3. Centrifuge at 16,000 g for 1 min.
4. Discard the supernatant.
5. Repeat steps 2-4 until collected all medium.
6. Add 200 µL MX1 buffer.
7. Resuspend the pellet by vortexing or pipetting.
8. The cells should completely resuspend.
9. Add 250 µL MX2 buffer.
10. Gently mix the buffer well.
11. Do not vortex. Invert microtube 8-10 times.
12. Incubate the mixture at room temperature for 2 min.
13. Add 350 µL MX3 buffer.
14. Immediately and gently mix the solution. (Do not vortex.)
15. Centrifuge at 16,000 g for 5 min.
16. While centrifuging, assemble the column and collection tube.
17. Transfer the supernatant, around 500 µL (take 800 µL next time), to the column.
18. Centrifuge at 7,000 g for 30 sec.
19. Discard the flow through.
20. Add 500 µL WN buffer.
21. Centrifuge at 7,000 g for 30 sec.
22. Discard the flow through.
23. Add 700 µL WS buffer.
24. Centrifuge at 7,000 g for 30 sec.
25. Discard the flow-through.
26. Centrifuge the empty column at 10,000 g for 3 min to remove residual ethanol.
27. Place the column into a new 1.5 mL microtube.
28. Add 50 µL sterile distilled water on the center of the membrane.
29. Incubate the column at room temperature for 3 min.
30. Centrifuge at 10,000 g for 3 min.

Results:

Recorded from biodrop.

	Concentration	A2601230	A2601280
A	201 $\mu\text{L/mL}$	1.305	1.990
B	211 $\mu\text{L/mL}$	1.294	1.991
C	187 $\mu\text{L/mL}$	1.127	1.948
D	119 $\mu\text{L/mL}$	0.922	1.859
E	195 $\mu\text{L/mL}$	1.373	2.010

Done by Sophie L, Noah, Anton.

Supervised by: Gabriel.

ELN by Sophia Y.

Date: 20260605

Title: Prepare 0.8% Agarose Gel for DNA Electrophoresis

Purpose: Do DNA electrophoresis for PCR product

Procedure:

1. Assemble the mold and tray, and prepare the comb.
2. Weigh 0.16 agarose powder and obtain 20 mL 1X TAE solution.
3. Add 1X TAE and powder to the conical flask.
4. Cover the flask with plastic wrap, then use a tip to puncture a hole in the plastic wrap.
5. Microwave for 30 sec. Then take it out and agitate it.
6. Microwave for 10 sec and agitate it. Repeat this step until there is no particle in melted gel.
7. Wait until the melted gel stops smoking but is still hot.
8. Add 1 μL safety dye and agitate it.
9. Pour the melted gel into the tray. Then assemble the comb.

Done by Sophie L

Supervised: Chars, Gabriel

ELN by Sophia Y.

Date: 20260605

Title: LB Medium Preparation

Purpose: prepare LB Medium

Procedure:

1. Formula as below:

LB broth powder	12.5g
ddH ₂ O	500mL

2. Add one magnetic stirring rod to the flask.
3. Place the flask containing the mixture on the hot plate and stir until dissolved.
4. Remove the stirring rod from the flask.
5. Cover the flask with aluminum foil and apply an indicator tape.
6. Place the flask in an autoclave.

Done by Abby T and Abigail.

Supervised by Chars and Gabriel.

ELN by Sophia Y.

Date: 20260605

Title: Plasmid linearization by enzyme digestion

Purpose: to prepare the plasmid for gel electrophoresis

Procedure:

A: pTWIST-Amp-SamyQ-AmiCO

B: pTWIST-Amp-H01-PcyA

C: pTWIST-Amp-CcaSm3

D: pTWIST-Amp-CcaR

E: pTWIST-Amp-Csn:LEA

1. Calculate reaction formula:

	A	B	C	D	E
10X cut smart buffer	2µL	2µL	2µL	2µL	2µL
EcoR1-stF	0.5µL	0.5µL	0.5µL	0.5µL	0.5µL
DNA	2.5µL	2.4µL	2.7µL	4.2µL	2.6µL
ddH ₂ O	15µL	15.1µL	14.8µL	13.3µL	14.9µL

2. Prepare the reaction
 - First add water DNA and buffer
 - Then add enzymes
3. Incubate in PCR machine

Done by Chloe

2026/6/6



Date: 20260606

Title: Golden Gate Assembly

Purpose: directionally assemble a promoter, RBS, coding sequence, and terminator into a destination vector in a single reaction.

Procedure:

1. Assemble the following:

a.

Vector (pJUMP29-1A): Dilute 1:10, use 2.48 μL (1 μL pJUMP + 8.1 μL ddH₂O).

Insert Promoter (pcpG2-veg): Dilute 1:100, use 1.87 μL (1 μL pcpG2-veg + 99 μL ddH₂O).

Insert RBS (EEBD-m1001): Dilute 1:100, use 3.28 μL (1 μL EEBD RBS + 99 μL ddH₂O).

Insert CDS (pTwist-Amp-samyc): Dilute 1:10, use 3.95 μL (1 μL AmiCP + 14.1 μL ddH₂O).

Insert Terminator (B0015): Dilute 1:10, use 3.28 μL (1 μL B0015 + 5.6 μL ddH₂O).

T4 DNA Ligase Buffer: 2 μL .

Sterile Water: Add to a final volume of 20 μL (1.14 μL).

T4 DNA Ligase: 1 μL .

Bsal-HFv2: 1 μL .

b.

Vector (pJUMP29-1A): Dilute 1:10, use 2.48 μL (1 μL pJUMP + 8.1 μL ddH₂O).

Insert Promoter (523100): Dilute 1:100, use 4.6 μL (1 μL promoter stock + 99 μL ddH₂O).

Insert RBS (EEBD-m1001): Dilute 1:100, use 3.28 μL (1 μL EEBD RBS + 99 μL ddH₂O).

Insert CDS (pTwist-Amp-samyc): Dilute 1:10, use 3.95 μL (1 μL AmiCP + 14.1 μL ddH₂O).

Insert Terminator (B0015): Dilute 1:10, use 3.28 μL (1 μL B0015 + 5.6 μL ddH₂O).

T4 DNA Ligase Buffer: 2.6 μL .

Sterile Water: Add to a final volume of 23 μL (0.54 μL).

T4 DNA Ligase: 1.27 μL .

Bsal-HFv2: 1 μL .

2. Incubate both assembly mixtures in a PCR machine using the appropriate Golden Gate Assembly program.

Done by: Chloe, Sophie L., and Ryan Y.

Supervised by: Hachi.

Date: 20260606

Title: *Bacillus subtilis* growth rate in bioreactor

Procedure: test the rate *Bacillus subtilis* growth in bioreactor

1. Measure OD600 *Bacillus subtilis* growth rate in bioreactor and incubator every 30mins

Results:

	Culture in bioreactor	Culture in incubator
0 min	0.172	0.172
30 min	0.108	0.189

60 min	0.101	0.205
90 min	0.194	0.217
120 min	0.130	0.230
150 min	0.178	0.347
180 min	0.166	0.314
210min	0.112	0.309
240 min	0.123	0.377
270 min	0.136	0.442
300 min	0.126	0.556
330 min	0.125	0.698
360 min	0.083	0.811
390 min	0.131	0.987

Done by Joshua, Chloe, and Ryan W.
ELN by Sophia Y.

Date: 20260606

Title: Liquid Culture of pJUMP1A–pJUMP29-1D

Purpose: grow and amplify bacterial cells from single colonies

Procedure:

1. Prepare TB with antibiotics (Master Mix): 40 mL TB broth + 40 μ L kanamycin (50 mg/mL).
2. Split the master mix into 8 tubes, each containing 5 mL.
3. Inoculate one colony from the glycerol stock into each tube.
4. Place the tubes into the incubator.

Done by: Joshua, Ryan Wei, and Hachi.

Date: 20260606

Title: Electrophoresis

Purpose: Verify the gene synthesis produced.

Procedure:

A: pTwist-Amp-SamyQ:AmilCP

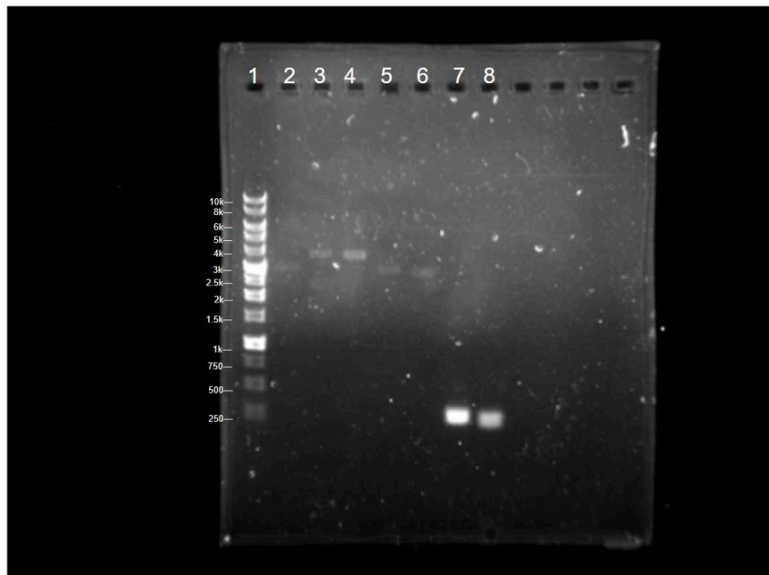
B: pTwist-Amp-Ho1:pcyA

C: pTwist-Amp-ccaSm3
D: pTwist-Amp-ccaR
E: pTwist-Amp-csn:LEA

1. Mix 5 μ L DNA with 1 μ L loading dye.
2. Loading sample by the following layout:

M	A	B	C	D	E	EEBD	PcpcG2-veg
5	6	6	6	6	6	6	6
3. Resolve the DNA by 100 V 30 min.

Results:



1: Omics 1kb DNA RTU Ladder 5 uL
2: pTwist-Amp-High-SamyQ:AmilCP 1hr 5 uL (1.005 ug)
3: pTwist-Amp-High-Ho1:pcyA 1hr 5 uL (1.055 ug)
4: pTwist-Amp-High-ccaSm3 1hr 5 uL (935 ng)
5: pTwist-Amp-High-ccaR 1hr 5 uL (595 ng)
6: pTwist-Amp-High-Csn:Lea 1hr 5 uL (975 ng)
7: EEBD
8: Pveg (237bp)

Done by Sophie L, Sophie C, and Sarah C.
Supervised by Hachi.
ELN by Sophia Y.

2026/6/7



Date: 2026.06.07

Title: Chlorophyll Extraction

Purpose: Extract chlorophyll to evaluate how well our arabidopsis is growing.

Procedure:

1. Weigh the fresh weight (fw) of arabidopsis (10 at a time).
2. Take the plant from the agar plate, careful not to hurt it by squeezing the tweezers.
3. Put the 10 plants on a tissue paper, lightly press to remove the agar sticking to the plants.
4. Pick up the plants and put them in a microtube, take care that the roots need to go in, too.
5. Weigh the plants

Repeat these steps for as many times as the sets of arabidopsis you are testing.

6. Place 3 glass bits in each of the microtubes, place in -80°C for 10 minutes.
7. Dilute acetone to 80%, add 1 mL to each tube.
8. Wait for the chlorophyll to extract, and can use a vortex to crush the plants to speed up the process (done when plants are no longer green.)
9. Add the liquid to the circus 96 well, 200 µL in each well, 3 wells for each sample.
10. ELISA reader, insert the 96 well, run the results through the equation below.

*Total chlorophyll content (mg/g)

$(20.2 \cdot OD_{645} + 8.02 \cdot OD_{663}) \cdot V / 1000 \cdot fw$

Results:

	Well 1	Well 2	Well 3
E (0.0164)	451.83	479.27	739.02
F (0.0208)	348.56	390.87	391.35
G (0.0207)	346.38	384.54	390.34
H (0.0164)	428.05	473.17	485.98

Done by Phoebe C., Sophia Y., recorded by Phoebe C.

ELN by Sophia Y and Phoebe C.

Date: 2026.06.07

Title: Transformation of Bba_J232119 and Golden Gate Assembly of 6/6

Purpose: To transform competent cells with the Bba_J23119 plasmid and the completed Golden Gate Assembly construct and to isolate cells carrying the desired DNA constructs through antibiotic selection.

Procedure:

1. Turn on the water bath and set at 42 deg C, and prepare the ice.
2. Spread antibiotics on the agar plate.
3. Thaw the competent cell and aliquot 30 µL of the competent cell in each microtube.

4. Add 2 μL of plasmid to the 30 μL competent cell (do not pipette). 2 μL for J23119, 10 μL for J23106 and pepcG2-Veg.
5. Place on ice for 20 minutes.
6. Heat shock at 42°C for 45 seconds to 1 minute.
7. Place on ice for 3 minutes to recover, add 1 mL SOC medium, and incubate at 37°C for 1 hour.
8. Centrifuge at 5000 g for 5 minutes and remove 1 mL of supernatant, then resuspend pellet.
9. Spread the competent cell on agar plate with kanamycin.
10. Incubate the plate at 37°C for 16 hours.

Done by Chloe, Phoebe, and Sophia Y, recorded by Chloe.

ELN by Sophia Y.

Date: 2026.06.07

Title: Agar Preparation

Purpose: Prepare agar solution

Procedure:

1. Formula as below:
 2.37 g $\frac{1}{2}$ MS Medium
 0.276 g $\frac{1}{2}$ MS Medium (no sucrose)
 180 mL ddH₂O
 1.386 g agar
2. Add contents into a beaker.
3. Use the hot plate to stir (with a stirring rod) until no sediments are left in the solution.
4. Adjust the pH with 0.1 M KOH to 5.7-5.8.
 Original pH: 5.40
 Final pH: 5.74
5. Wrap the neck with aluminum foil and place indicator tape on top.
6. Autoclave for 20 min at 127°C.
7. Wait for the pressure to reach 0 before retrieving the flask.

Done by Sophia Y and Chloe, recorded by Sophia Y.

ELN by Sophia Y.

Date: 20200607

Title: *Bacillus subtilis* growth rate in Bioreactor (Continuous from 20260606)

Purpose: continue 20260606 experiment

Procedure:

Time(hr)	Incubator	Bioreactor
24	2.169	0.197

25	2.461	0.342
26	2.402	0.370
27	-	0.304

Sampling from medium : 0.480

Done by Chars

2026/6/8



Date: 20260608

Title: Agar Preparation

Purpose: prepare agar

Procedure:

1. Formula as below:

½ MS medium(with sucrose)	2.37 g
½ MS medium(without sucrose)	0.276 g
Agar	1.386 g
ddH ₂ O	180 mL

2. Add contents into a flask
3. Use the hot plate to stir(with a stirring rod) until no sediments are left inside the solution
4. Adjust the pH using a pH probe to around 5.7 using 0.1M KOH
 - a. Record the original pH and final pH values
5. Wrap the neck with aluminum foil and place the indicator tape on top
6. Autoclave for 20 minutes at 127°C
7. Wait for pressure to reach 0 before opening the autoclave

Done by: Abigail and Sophia Y.

Supervised by Gabriel.

ELN by Sophia Y.

Date: 2026.06.08

Title: Prepare 1.0% Agarose Gel for DNA electrophoresis

Purpose: Prepare 1.0% Agarose Gel for DNA electrophoresis

Procedure:

1. Assemble the mold and tray, and prepare the comb.
2. Weigh 0.2 g of agarose powder and obtain 20 mL 1X TAE solution
3. Add 1X TAE and agarose powder to the conical flask.
4. Cover the flask with plastic wrap, then use a tip to puncture holes in the plastic wrap
5. Microwave for 30 s and agitate it. Repeat this step until there are no particles in the melted gel.
6. Wait until the gel stops smoking but is still hot.
7. Add 1 µL of safety dye into the gel and agitate it.
8. Pour the melted gel into the tray, then assemble the comb.

Done by Naomi, recorded by Naomi.

Supervised by Gabriel.

ELN by Sophia Y.

Date: 20260608

Title: Colony PCR

Purpose: amplify golden assembly parts(6/7) with 2 different promoters

Procedure:

1. Prepare the master mix(9 reactions)

The following is the formula for 1 reaction:

Forward Primer(Biobrick prefix)	1
Reverse Primer(Biobrick suffix)	1
Q-Amp Master Mix	10
d2W	8

2. Dispense reaction to 8 strip PCR tube, each 4 colonies for each of the two dishes
3. Pick up colonies and tap 3 times on a separate plate for each of the colonies to incubate
4. Release the tips with the colonies in the PCR tubes and wait for 5 minutes
5. Take out the tips and put tubes in PCR machine
6. Thermal cycle

95°C	5min	1cycle
95°C	30sec	35cycle
55°C	30sec	
72°C	1min	
72°C	5min	1cycle

Done by: Abigail and Sophia Y.

Supervised by: Gabriel.

ELN by Sophia Y.

Date: 20260608

Title: Hydroponic Medium Preparation

Purpose: prepare hydroponic medium

Procedure:

1. Measure 2.07g of ½ MS medium in 900mL of de-ionized water
2. Use the hot plate to stir(with a stirring rod) until no sediments are left in the solution
3. Adjust the PH using a plt probe to around 5.7-5.8 use 0.1 M KOH
Original pH : 4.00
Final pH:5.1

4. Split into two beakers (by estimation)
5. Wrap the neck with aluminum foil and place the indicator tape on top
6. Autoclave for 20 min at 127°C
7. Wait for the pressure to reach 0 before opening the autoclave

Done by Sophia Y and Abigail

Supervised by Gabriel

Date: 20260608

Title: Sanitizing equipment

Purpose:disinfect laboratory tools using a 10% bleach solution.

Procedure:

1. Pour 200mL of pure bleach into a graduated cylinder then into a 2 L beaker
2. Add 1.8 L of dd H₂O to the beaker
3. Pour 10 % bleach into a tub and fully cover the equipment in the tub
4. Wait for 10 minutes
5. Rise the equipment ddH₂O until these is no bleach sent

Done by Abigail and Sophia Y

Supervised by Gabriel

2026/6/9

Date: 20260609

Title: Prepare 1.0% Agarose Gel for DNA electrophoresis

Purpose: Do DNA electrophoresis for colony PCR product(6/8)

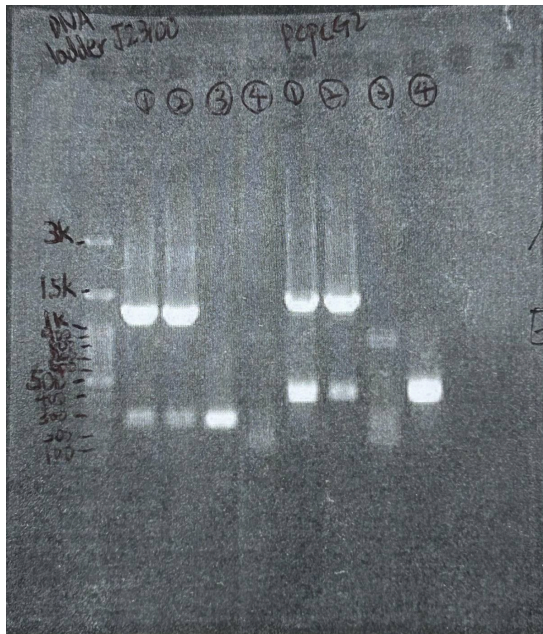
Procedure:

1. Assemble the mold and tray, prepare the comb
2. Weigh 0.2g of agarose powder and add it into a flask
3. Add 20ml of 1xTAE solution into the flask
4. Cover the flask with plastic wrap then use a tip to puncture holes on the plastic wrap
5. Microwave for 30 seconds and agitate it
 - a. Repeat this step until all dissolved and no particles left in the mixture
6. Wait until the gel stops smoking but is still hot
7. Add 1 microliter of safety dye into the gel and agitate it
8. Pour the melted gel into the tray and then assemble the comb
9. Wait for the gel to cool

Done by Abby K, Sophia Yeh, Abby T

Supervised by Gabriel

Results:



Date: 20260609

Title: Miniprep

Purpose: to amplify P1~4, J1~4

Procedure:

1. Transfer 1 mL TB medium to 1.5mL microtube

- a. PCPCG2 sample 1
- b. PCPCG2 sample 2
- c. PCPCG2 sample 3
- d. PCPCG2 sample 4
- e. J23100 sample 1
- f. J23100 sample 2
- g. J23100 sample 3
- h. J23100 sample 4
- i. J23119(promoter)

*vector is pJUMP29-1A

- 2. Centrifuge at 16,000 g for 1 minute
- 3. Discard the supernatant
- 4. Repeat steps 1-3 until collected all medium
- 5. Add 200microliters Mx1 buffer
- 6. Resuspend the pellet by vortexing or pipetting
- 7. The cells should completely resuspend
- 8. Add 250 microliters of Mx2 buffer
- 9. Gently mix buffers well
- 10. Do not vortex. Invert microtube 8-10 times
- 11. Incubate the mixture at room temperature for 2 minutes
- 12. Add 350 microliters Mx3 buffer
- 13. Immediately and gently mix the solution(do not vortex)
- 14. Centrifuge at 16,000g for 5 minutes
- 15. While centrifuging, assemble the column and collecting tube
- 16. Transfer the supernatant, around 800 microliters to the column
- 17. Centrifuge at 7,000g for 30 seconds
- 18. Discard the flow through
- 19. Add 500 microliters WN buffer
- 20. Centrifuge at 7000g for 30 seconds
- 21. Discard the flow through
- 22. Add 700 microliters of WS buffer
- 23. Centrifuge at 7,000g for 30 seconds
- 24. Discard the flow through
- 25. Centrifuge the empty column at 10,000 g for 3 min to remove the residual ethanol
- 26. Place the column into a new 1.5ml microtube
- 27. Add 50 microliters of sterile distilled water on the center of the membrane
- 28. Incubate the column at room temperature for 3 minutes
- 29. Centrifuge at 10,000g for 3 minutes

Done by: Abigail, Abby K, Jacquelyn, Sophia Y
Supervised by Gabriel

Date: 20260609

Title: Enzyme Digestion of J2 (J23100 sample 2)

Purpose: cut DNA molecules at specific sequence sites

Procedure:

1. Calculate reaction formula

- Double restriction

	J23110(J1)	pNW33N-L2 (sfGFP) (S1)
10X r2.1 buffer	5 μ L	5 μ L
EcorI-HF	1 μ L	1 μ L
pstI	1 μ L	1 μ L
DNA	9.8 μ L	13 μ L
ddH ₂ O	33.2 μ L	30 μ L

- linearization(single)

	J23110(J3)	J23110(J2)
10X r2.1 cut smart buffer	2 μ L	2 μ L
EcorI-HF	0.5 μ L	0.5 μ L
DNA	4.6 μ L	4.9 μ L
ddH ₂ O	12.9 μ L	12.6 μ L

2. Prepare the reaction

- a. First add water, DNA, and buffer
- b. Then add enzyme

(Prepare double restriction twice)

3. Incubate in PCR machine

Done by: Chloe, Sophie L, Abby T, Abigail

Supervised by Gabriel

Date: 20260609

Title: Prepare 1.0% agarose gel for electrophoresis

Purpose: prepare gel for electrophoresis

Procedure:

1. Assemble the mold and tray, and prepare the comb
2. Weigh 0.2g of agarose powder and obtain 20mL 1X TAE solution
3. Add agarose powder and 1X TAE to a conical flask
4. Cover the flask with plastic wrap , then use a tip to puncture holes in the plastic wrap
5. Microwave for 30sec. Then take it out and agitate
6. Microwave for 10 sec and agitate it. Repeat this step until there is no particle in melted gel

7. Add 1 μ L of safety dye into the gel and agitate

8. Pour the melted gel into the tray. Then assemble the comb

Done by Sophie L

Supervised by Gabriel

2026/6/10

Date: 20260610

Title: Prepare 1.0% Agarose Gel for Electrophoresis

Purpose: Preparation for separating DNA fragments

Procedure:

1. Assemble the mold and tray.
2. Weight 0.2g of agarose powder and add it into a flask.
3. Add 20 ml of 1x of TAE solution into the flask.
4. Cover the flask with plastic wrap and puncture some holes.
5. Microwave for 30 seconds and agitate it. Repeat this step until all particles are dissolved.
6. Wait until the gel stops smoking but is still hot.
7. Add 1 μ L of safety dye into the gel and agitate it.
8. Pour the melted gel into the tray then assemble the comb.

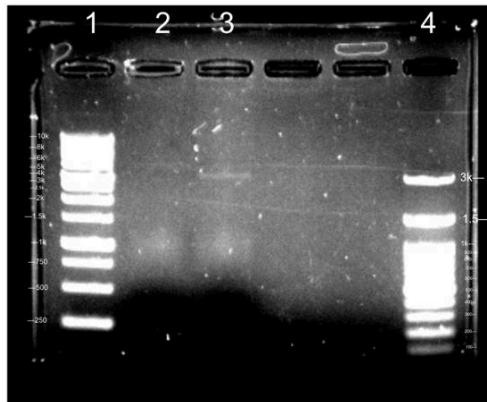
Done by Abby T and Ethan Chang

Supervised by Chars

Recorded by Audrey H.

Results:

260610 EcoR1



1. Omics 1kb DNA RTU Ladder
2. J23100 sample 2(highest A230/260)
3. J23100 sample 3(highest A230/260)
4. [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL

Date: 20260610

Title: Electrophoresis

Purpose: separate DNA fragments

Procedure:

1. Mix 4 μ L of safety dye with DNA.

2. Loading sample by the following layout.

M	A	B		C	
5				5	volume (μL)
M= 1kb DNA ladder (blue) A= J2		B= J3 C= 1 kb DNA ladder (green)			

3. Resolve the DNA by 100V 30 minutes.

Done by Abby T

Supervised by Chars

Recorded by Audrey H.

Date: 20260610

Title: DNA Ligation

Purpose: Insert the J23100 driven Csn:AmilCP into pNW33N-L2 (sfGFP)

Procedure:

1. Calculate the Insert-Vector Ratio for DNA Ligation based on the molar ratio of 1:3 and formula as below:
2. The ratio as below:

Vector : Insert
1 : 3.58

J23100	3.9 pmol
pNW33N-L2 (sfGFP)	4.3 pmol

3. Prepare the reaction by the following table:

	Volume (μL)
10X T4 DNA Ligase Reaction Buffer	2 μL
pNW33N-L2 (sfGFP)	4.3μL
J23100	3.9μL
Nuclease- Free water	8.8μL
T4 DNA Ligase	1μL

4. Leave for 2 minutes.

Done by Jacquelyn, Abby T and Abigail

Supervised by Chars

Recorded by Audrey H.

Date: 20260610

Title: Electrophoresis and Gel extraction

Purpose: Isolate desired DNA fragment from enzyme digestion

Procedure:

1. Mix 50 μ L DNA with 10 μ L 6X loading dye.
2. Loading sample by the following layout.

M	A	B	C	
5	60	60	5	volume (μ L)
M= 1kb DNA ladder (blue)		B= J1 DR		
A= S1		C= 1 kb DNA ladder (green)		

3. Resolve the DNA by 100V 30 minutes.

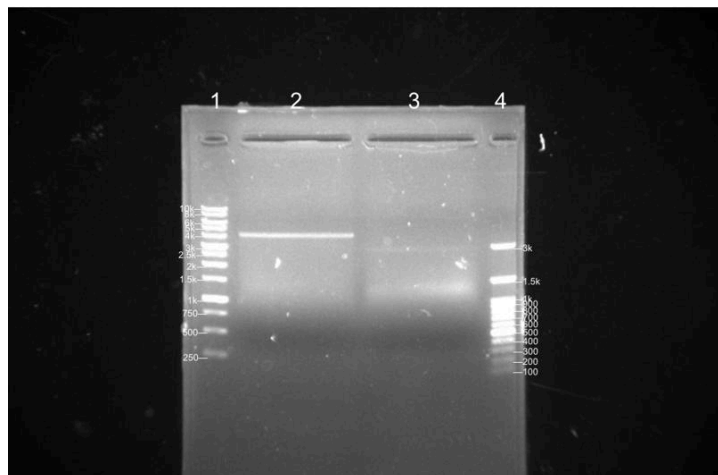
Done by Ethan Chang

Supervised by Chars

Recorded by Audrey H.

Results:

260610 Gel extraction



- 1: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
- 2: pNW33N-L2(sfGFP) EcoRI PstI 16 hr
- 3: J23100 Transcript unit EcoRI PstI 16 hr
- 4: Omics 1kb DNA RTU Ladder 5 uL

Date: 20260610

Title: Gel Extraction

Purpose: DNA fragment isolation and purification.

Procedure:

1. Use a sanitized sharp knife to cut the gel band containing the desired DNA fragment under the UV light box.
2. Place each 2 gel bands into separated and labeled 2ml tubes (pNW33N-L2 (sfGFP) EcoRI-PstI and J23100 EcoRI-Pst1).
3. Add 700 μ L of GP Buffer into each of the tubes.
4. Place the tubes in a dry bath to dissolve the gel slices, check every 2 minutes until entirely dissolved.
5. Centrifuge for 30seconds. Discard the flow-through. Repeat this step for the rest of the mixture.
6. Wash the column once with 0.5ml of WN Buffer by centrifuging for 30 seconds. Discard the flow through.
7. Wash the column once with 0.5ml of WN Buffer by centrifuging for 30 seconds. Discard the flow through.
8. Centrifuge the column at full speed 23444g for 3 minutes to remove residual ethanol.
9. Place the column onto a new centrifuge tube. Add 15 μ L of Nucleus free water that is heated by the thermal shaker onto the center of the membrane.
10. Stand the column for 2 minutes and centrifuge at full speed 23444g for 2 minutes to elute DNA.
11. Turn on BioDrop, adjust the setting to pathlength unit (concentration).
12. Clean the sample port with 1 μ L of Nucleus free water.
13. Calibrate.
14. Load 1 μ L of DNA from the 2 tubes each time and record down the results.
15. Clean the port with Nucleus Free water and close BioDrop.

Done by Abby T

Supervised by Chars

Recorded by Audrey H

Date: 20260610

Title: Transformation of pNW33N-L2 (sfGFP)

Purpose: Bacterial transformation with plasmid DNA.

Procedure:

1. Turn on the water bath and set it to 42°C and prepare ice.
2. Gel agar plate with antibiotics (Chloramphenicol).
3. Thaw 100 μ L of the competent cell and add 10 μ L plasmid (don't pipette).
4. Place on ice for 10 minutes.
5. Heat shock at 42°C for 45 to 60 seconds.
6. Place on the ice for 5 minutes to recover and add 1 mL SOC medium, then incubate for 1 hour at 37°C.
7. Centrifuge at 16,000 g for 30 seconds, remove 1 mL supernatant and resuspend pellet.
8. Spread competent cells on agar plate with Chloramphenicol .
9. Incubate at 37°C for 16 hours.

Done by Jacquelyn, Abby T, Chloe, Abigail, Sophie L, and Olivia L
Supervised by Chars
Recorded by Audrey H

2026/6/11



Date: 20260611

Title: Agar Prep

Purpose: make 1% sucrose agar plates in rectangular form (2 plates)

Procedure:

1. Formula as below:
 - sucrose: 1.0 g
 - w/o sucrose 1/2 MS medium: 0.23 g
 - agar: 1.0 g
 - MES: 0.05 g
2. Add contents into a beaker with ddH₂O
3. Stir with a stirring rod on a hot plate until no sediments are left in the solution.
4. Add KOH & HCl to adjust the pH to 5.7 ~ 5.8.
 - original pH: 4.1
 - final pH: 5.42
5. Pour contents into a graduated cylinder.
6. Fill with ddH₂O up to 100 mL.
7. Pour contents back to flask (shake to ensure all contents are poured).
8. Cover flask with aluminum foil & place indicator tape on top.
9. Autoclave for 20 minutes at 127°C.
10. Pour the agar mixture into the rectangular plates and wait for it to solidify.

Done by: Abigail, Jacquelyn, Sophie C, Joshua

Supervised by: Chris

2026/6/12



Date: 20260612

Title: Transformation PcpG2-veg/EEBD-MF001

Purpose: Bacterial transformation with plasmid DNA.

Procedure:

1. Turn on the water bath, set it to 42°C and prepare the ice.
2. Mix 100 mL competent cell with 2 mL plasmid (don't pipette)
3. Incubate on ice for 20 minutes.
4. Heat shock at 42°C for 45 seconds.
5. Incubate on ice for 3 minutes.
6. Put in incubator at 37°C for 10 minutes.
7. Centrifuge 3000 g for 5 minutes.
8. Taking out 1000 ML and resuspend the cell.
9. Spread on the agar plate with Ampicillin.
10. Place it in incubator for 37°C overnight.

Done by Sarah C.

Supervised by Hachi

Recorded by Sophie Chen, Audrey H

Date: 20260612

Title: TA cloning

Purpose: Create plasmid by cloning insert into vector

Procedure:

1. Calculate the insert volume.
Ratio: 3:1
 - PcpG2-veg : $0.01363(\mu\text{L}) / 0.168(\mu\text{g/mL}) = 0.072$
 - EEBD-MF001: $0.01204 / 0.109 = 0.125$
2. Dilute the insert PcpG2-veg 20X and EEBD-MF001 10X

Reagents	Standard Reaction (ML)	
2X Rapid ligation Buffer, T4 DNA Ligase	5	5
pGEM-T Easy Vector (50ng) (3015bp)	1	1
T4 DNA Ligase	1	1
Denionized water to final volume of	10	10
insert PcpG2-veg (242bp)	1.44	
insert EEBD-MF001 (274 bp)		1.25

3. Mix the reaction by pipetting.
4. Incubate the reaction one hour at room temperature.

Done by Sarah C.

Supervised by Gabriel

Recorded by Audrey H

2026/6/13 

Date: 20260613

Title: Colony PCR for PCPC62-veg / EEBD-MF001's transformation 0612

Purpose: verify if the bacterial colonies successfully received the PCPC62-veg or EEBD-MF001 transformation inserts.

Procedure:

1. Draw a grid on new LB plate with Ampicillin and prewarm in 37°C (4 x 4 grid)
2. Prepare PCR reaction master mix in 1.5 mL microtube

2x Taq master mix	180 µL
Prefix	18 µL
Suffix	18 µL
PCR- grade water	144 µL

3. Aliquot mix to two 8-strip PCR tubes
4. Label the colonies numerically 1 to 8 (1 for EEBD & 1 for PCP)
5. Touch the single, well isolated colony gently using a sterile pipette tip
6. Streak to the labelled grid & Eject the tip into labelled PCR tube
7. Wait for 2 minutes and remove the tips and place reaction in PCR machine for the following times

Cycle	Step	Temperature (*C)	Duration
1	Denature	95	5 minutes
35	1. Denature 2. Annealling 3. Elongation	1. 95 2. 55 3. 72	4. 30 seconds 5. 40 seconds 6. 1 minute/Kb
1	Elongation	72	5 minutes

Done by Olivia Lin

Recorded by Audrey H

Date: 20260613

Title: LB Medium Reparation

Purpose: prepare LB medium

Procedure:

1. Formula as below (mixture for each of 2 flasks):

LB Broth Powder	12.5g
ddH ₂ O	500ml

2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate and stir until dissolved
4. Remove the stirring bar from the flask

5. Cover the flask with aluminum foil and apply the indicator tape
6. Autoclave for 20 minutes at 127°C

Done by Abby T, Sara Chen

Supervised by Chars

Recorded by Audrey H

2026/6/14



Date: 20260614

Title: ½ MS medium agar prep

Purpose: Arabidopsis salinity growth test

Procedure:

1. Prepare the following in a flask:

½ MS medium powder	1.15 g
MES buffer	0.25 g
Agar powder	5 g
Sucrose	5 g
Distilled water	500 ml

2. Add half of the water in and use a hot plate to stir until no more sediment
3. Add the remaining water and stir
4. Make sure the pH is between 5.6 to 5.7

Current pH: 4.68  After adding 0.1M KOH: 5.7

5. Transfer 250ml of the mixture into another flask.
6. Add 1.461 g of NaCl into a flask
 - a. Label each flask clearly
7. Separate both the NaCl and control mixtures into 4 separate flasks.
8. Label clearly each flask and their respective trehalose concentrations
 - a. 0mM, 0.25mM, 0.5mM, and 1mM
9. Add their respective trehalose amounts:

Concentration	100mM trehalose
0 mM	0 MI
0.25 mM	125 MI
0.5 mM	250MI
1 mM	500MI

10. Cover with aluminium foil and label with indicator tape.
11. Autoclave at 127°C for 20 minutes.
12. Pour into square petri dishes under the BSC (around 50ml each) using a pipette gun
13. Let solidify under room temperature

Done by Abby K, Abigail

Recorded by Audrey H

Date: 20260614

Title: Bioreactor growth test

Purpose: measure the *Bacillus subtilis* 169 grow in the bioreactor

Procedure:

1. Assemble the bioreactor
2. Add 20 mL on culture medium to bioreactor & flask (volume 500mL)
3. Sample and measure OD600 culture in 37°C , 200rpm orbital shaker (flask) and 200 rpm stir (bioreactor)

Hr	Time	Flask	Bioreactor
0	11:00	0.122	0.162
0.25	11:15	0.130	0.148
0.5	11:42	0.154	0.137
0.75	12:02	0.173	0.176
1	12:19	0.226	0.262
	12:36	0.226	0.391
	12:53	0.257	0.517
	13:10	0.298	0.767
	13:25	0.380	0.895
	13:46	0.408	0.784
	2:05	0.439	0.831
	14:22	0.583	0.961
	14:37	0.537	0.875
	14:54	0.606	0.938
	15:10	0.640	0.879
	15:28	0.761	0.883
	15:43	0.832	0.819
	16:00	0.634	0.841
	16:19	0.622	0.741
	16:41	0.572	0.764
	16:58	0.686	0.916

	17:15	0.674	0.826
	17:32	0.657	0.812
	17:51	0.817	0.817
	18:10	0.732	0.843

Date: 20260614

Title: 1.0 Agarose

Purpose: prepare gel for electrophoresis

Procedure:

1. Assemble mold+tray:prepare the comb
2. Mix 0.4g agarose and 40 mL 1XTAE solution in a conical flask
3. Cover flask with plastic wrap and puncture holes
4. Microwave in short periods(30 sec then 10 sec) until powder is all melted
5. Add 2 μ L of safety dye into mixture and agitate
6. Pour mixture into the tray and assemble comb

Done by Joshu W+Phoebe

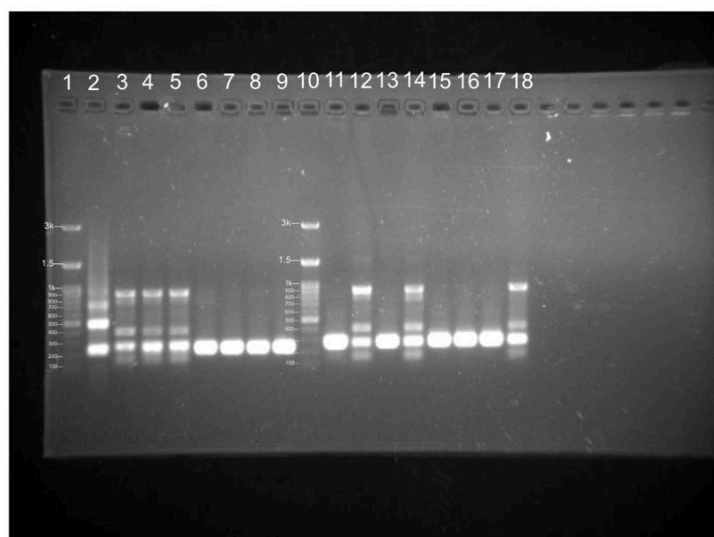
Date: 20260614

Title: Gel electrophoresis for PcpcG2-veg and EEBD-MF001

Purpose: test colony PCR result

Procedure:

1. Loaded (Colony PCR product of 20260613 Done by Olivia) with 1~8 ladder (PcpcG2-veg) and 9~16 with EEBD-MF001



1. [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
2. PCPCG2-veg sample 1
3. PCPCG2-veg sample 2
4. PCPCG2-veg sample 3
5. PCPCG2-veg sample 4
6. PCPCG2-veg sample 5
7. PCPCG2-veg sample 6
8. PCPCG2-veg sample 7
9. PCPCG2-veg sample 8
10. [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
11. EEBD-MF001 sample 1
12. EEBD-MF001 sample 2
13. EEBD-MF001 sample 3
14. EEBD-MF001 sample 4
15. EEBD-MF001 sample 5
16. EEBD-MF001 sample 6
17. EEBD-MF001 sample 7
18. EEBD-MF001 sample 8

Done by Joshua

2026/6/15 

Date: 20260615

Title: Miniprep

Purpose: Amplifying pcpcG2-veg and EEED-MP001

Procedure:

A,B: PcpcG2-veg

C,D: EEED-HF001

1. Transfer 1mL TB medium to 1.5mL microtube
2. Centrifuge at 16000g for 1 min
3. Discard the supernatant
4. Repeat steps 1-3 until collected all medium
5. Add 200µL MX1 buffer
6. Resuspend the pellet through vortexing or pipetting
7. The cells should be completely resuspend
8. Add 250µL of MX2 buffer
9. Gently mix the buffer well by inverting the microtube 8-10 times
10. Incubate the mixtures at RT for 2 min
11. Add 350µL MX3 buffer
12. Immediately and gently mix through invert 4-5 times
13. Centrifuge at 16000 g for 5 min
14. While centrifuging assemble the column and collection tube
15. Transfer the supernatant, around 800µL to the column
16. Centrifuge at 7000 g for 30sec
17. Discard the flow-through
18. Add 500µL WN buffer
19. Centrifuge at 7000 g for 30sec
20. Discard the flow-through
21. Add 700µL WS buffer
22. Centrifuge at 7000 g for 30 sec
23. Discard the flow-through
24. Centrifuge the empty column at 10000 g for 3 min to remove ethanol
25. Place the column into a new 1.5mL microtube
26. Add 50µL sterile distilled water on the center of the membrane
27. Incubate the column at RT for 3 min
28. Centrifuge at 10000 g for 3 min

Results:

The data below was measured with BioDrop.

	Concentration (ng/µL)	A260/280
A	93	1.898
B	90	1.957
C	168	1.953
D	74	1.897

Done by Abigail, Abby T, Abby K, and Audrey Hsieh.
Supervised by Gabriel Ng

Date: 20260615

Title: ½ MS Medium Preparation 700 mL

Purpose: Arabidopsis planting

Procedure:

1. Prepare the following in a large beaker
2. In two separate flasks, put 3.5g of agar powder in both
3. Mix the contents in the beaker using & hot plate until no sediments are left.
4. Adjust the pH until it is between 5.7-5.8 -> add 0.1M KOH to increase pH: before:3.92, after: 5.72
5. Pour 350mL of the mixture into each of the two flasks with agar powder
6. Use a stirring bar to stir until no sediments left
7. Cover with aluminium foil and stick indicator tape on top
8. Autoclave at 127°C for 20 minutes
9. Under the biosafety cabinet, use a pipette gun to separate the medium into 12 square petri dishes
10. Let the petri dishes solidify at room temperature

Done by Ethan Chang, Abby K, Sophia Yeh, Abby T, Audrey Hsieh

2026/6/17 

Date: 20260617

Title: ½ MS hydroponic box medium

Purpose: Make a total of 4 liters of hydroponic medium for hydroponic box

Procedure:

1. Measure 4.6g ½ MS medium(without sucrose) and 1g of MES in 2L ddH₂O water
 - a. Measuring formula $\Rightarrow (2.3\text{g/L}) \times 2 = 4.6$
2. Use the hot plate to stir(with a stirring rod) until no sediments are left in the solution
3. Determine pH of the solution and adjust the pH using a pH probe to around 5.6-5.7 using 1M KOH
 - a. Record original and final pH
4. Wrap the neck with aluminum foil and place the indicator tape on top
5. Autoclave at 127°C for 20 minutes
6. Wait for the pressure to reach 0 before opening the autoclave

Date: 20260617

Title: LB Agar

Purpose: Prepare LB Agar for further bacteria cultivation and transformation

Procedure:

1. Measure 6.25g of LB broth and 3.7g of agar in 250ml of distilled water
2. Use the bar plate to stir until no sediments remain in the solution
3. Autoclave at 127°C for 20 min
4. Bring to BSC to add 250µL of chloromphenicol into the autoclaved agar solution
5. Pipette the solution (10-15µL) into respective petri dishes and leave to dry (solidify)

Done by Abigail, Renee

Supervised by Chars

Record by Sophie H

Date: 20260617

Title: Transformation of pTwist-Amp-Hlgh SamyQ:TAT-HA-AmiICP

Purpose: To transform engineered

Procedure:

1. Add 15µL of bacteria and 1µL of plasmid in the microtube (don't pipette)
2. Place on ice for 20min
3. Heat shock at 42°C for 45sec
4. Place on the ice for 3min to recover
5. Spread the mixture onto the LBA plate
6. Place in the incubator for 25°C overnight

Done by Abigile

Supervised by Chars

Record by Sophie H.

2026/6/18 

Date: 20260618

Title: Miniprep

Purpose: Collecting miniprep from 20 STK kit parts

Procedure:

1. Label the following:

	name	concentration(ng/ μL)	A260/230	A260/280
A1	pSTK-1A-sfGFP	115.0	1.264	1.917
A2	pSTK-1B-sfGFP	151.0	1.027	1.888
A3	pSTK-1C-sfGFP	124.0	1.246	1.919
A4	pSTK-1D-sfGFP	132.0	1.179	1.913
A9	pSTK-Exp-1-sfGFP	162.0	1.141	1.884
C10	pSTK-Exp-2-sfGFP	103.0	1.493	1.981
D2	pSTK-0A-Pveg(BBa)	280.0	1.728	2.059
D3	pSTK-0A-PlepA	155.0	1.422	1.987
D4	pSTK-0A-PliaG	130.0	1.150	1.940
D5	pSTK-0A-PxylA	168.0	1.135	1.909
D7	pSTK-0A-Phyperspar	125.0	1.562	1.953
D8	pSTK-0A-BBa_Rodos	78.0 *	1.345	1.902
D9	pSTK-0a-BBa_R7800	122.0	1.173	1.902
D10	psTK-0AB-BtagC + RBS	125.0	1.645	1.984
D11	pSTK-0B-3P_promoer	106.0	1.093	1.828
E7	pSTK-B-RBS0	188.0	1.635	2.066
E10	pSTK-RS-TT-RS-Rib	150.0	1.210	1.974
G11	pSTK-0D-B0015	107.0	1.216	1.981
G12	pSTK-0D-L3S2P21	170.0	0.762	1.789
E2	pSTK-0C-eCFP_codon	142.0	1.406	1.972

2. Transfer 1.8 mL of TB medium to a 2 mL microtubule.
3. Centrifuge for 60 seconds at 16,000 g.
4. Discard the supernatant.

5. Repeat steps 2-3 until there is no TB medium left in the tube
6. Add 200 µl of MX1 buffer.
7. Resuspend the pellet through vortexing or pipetting.
8. The cells should be completely resuspended before moving to step 9.
9. Add 250 µl of MX2 buffer.
10. Gently mix the buffer well by inverting the microtube back and forth for 8-10 times.
11. Incubate the mixture at room temperature for 120 seconds.
12. Add 350 µl of MX3 buffer to the mixture.
13. Immediately and gently mix via inversion 4-5 times.
14. Centrifuge the mixture at max rcf [23,444 g] for 480 seconds.
15. While centrifuging, assemble the column and collection tube.
16. Transfer the supernatant (around 800 µl) to the column using a 200 µl micropipette to avoid transferring/contaminating the supernatant with the pellet.
17. Centrifuge the column at 7,000 g for 30 seconds.
18. Discard the flow-through liquid in the collection tube.
19. Add 500 µL WN buffer.
20. Centrifuge at 7,000 g for 30 seconds.
21. Discard the flow-through.
22. Add 700 µL WS buffer.
23. Centrifuge at 7,000 g for 30 seconds.
24. Discard the flow-through.
25. Centrifuge the empty column at 10,000 g for 3 minutes (next time 8 minutes,) to remove residual ethanol.
26. Place the column into a new 1.5 mL microtube.
27. Add 50 µL sterile distilled water on the center of the membrane.
28. Incubate the column at room temperature for 3 minutes.
29. Centrifuge at 10,000 g for 3 minutes.

Done by Audrey Hsieh, Sophia Yeh, Abby Kao, Ethan Chang
 Supervised by: Chars Hsieh and Gabriel Ng
 ELN by Ethan Chang

Date: 20260618

Title: Enzyme Digestion

Purpose: To test if enzymes have a reaction with the newly-miniprep DNA.

Procedure:

1. Prepare the mastermix:

rCutSmart (buffer)	40 µl
EcoR1	10 µl

ddH ₂ O	200 µl
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2. Place the contents into a 1.5 mL microtube (EcoR1 last).
3. Mix the contents via pipetting, and distribute 15 µl into 18 separate 0.5 mL microtube
4. Add 5 µl of DNA to the 0.5 mL microtube *[from 20260618 Miniprep]*
 - a. Add 5 µl of DNA to the tube and label it respectively to its STK name (pSTK...).
 - b. Repeat step 4a until all DNA has been added **except** C10 and A9 (Exp-2 sfGFP and Exp-1 sfGFP).
5. Incubate the labeled 0.5 mL microtubes in MiniAmp thermocycler at 37C indefinitely until the laboratory is ready to experiment a gel electrophoresis to size the STK DNA

Done by Sophia Yeh and Ethan Chang
 Supervised by Gabriel Ng
 ELN by Ethan Chang

Date: 20260618

Title: Electrophoresis

Purpose: Size the STK DNA

Procedure:

1. Mix 5 µl DNA with 1 µl 6X loading dye
2. Repeat step 1 until all 18 STK DNA has been loaded
3. Load the samples by the following layout (DNA/Volume in µl):

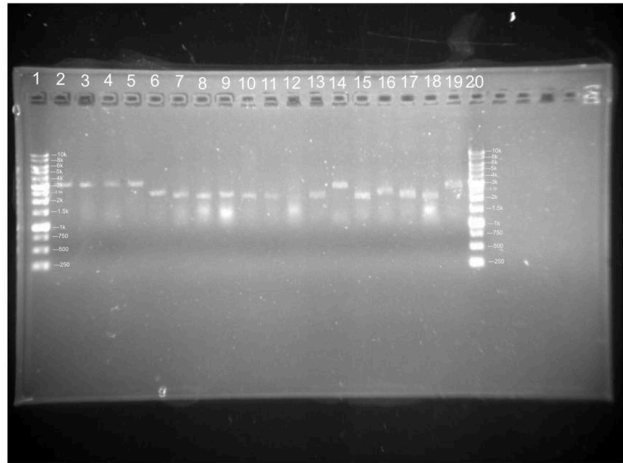
<u>M</u>	<u>A1</u>	<u>A2</u>	<u>A3</u>	<u>A4</u>	<u>D2</u>	<u>D3</u>	<u>D4</u>	<u>D5</u>	<u>D6</u>	<u>D7</u>	<u>D8</u>	<u>D9</u>	<u>D10</u>	<u>D11</u>	<u>E7</u>	<u>E10</u>	<u>E11</u>	<u>G12</u>	<u>F2</u>	<u>M</u>
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

M = 1Kb DNA ladder

4. Resolve the DNA by 100V for 30 minutes

Done by Abby Kao & Abby Tsai
 Supervised by Hachi Wu

20260618 STK plasmid



1. Omics 1kb DNA RTU Ladder
2. pSTK-1A-sfGFP
3. pSTK-1B-sfGFP
4. pSTK-1C-sfGFP
5. pSTK-1D-sfGFP
6. pSTK-0A-Pveg(BBa)
7. pSTK-0A-PlepA
8. pSTK-0A-PliaG
9. pSTK-0A-PxylA
10. pSTK-0A-Phyperspar
11. pSTK-0A-BBa K0905
12. pSTK-0A-BBa K7800
13. pStk-0AB-PtagC+RBS
14. pSTK-0B-3P promoter
15. pSTK0-B-RBS0
16. pSTK0B-RS-TT-RS-Rib
17. pSTK-0D-B0015
18. pSTK-0D-L3S2P21
19. pSTK-0C-eCFP codon
20. Omics 1kb DNA RTU Ladder

Date: 20260618

Title: making agarose gel 1%

Purpose: make 24 well agarose gel for gel electrophoresis

Procedure:

1. Prepare 0.4g of agarose powder and 40mL of 1X TAE in flask
2. Colder with piece of plastic wrap and use a pipette tip to poke 3-5 holes
3. Microwave per 30sec
4. Take it out to agitate
5. Microwave for 5 sec
6. Take out to agitate
7. Repeat steps 4-5 until all powder is dissolved
8. Remove plastic wrap and wait until warm but not too hot to touch
9. Add 2 μ L of safety dye, pipetting to mix
10. Pour into the rectangular model with 24 well comb
11. Use a pipette tip to press out air bubble
12. Leave to solidify at RT

Done by Abby K and Audrey

2026/6/19



Date: 20260619

Title: PCR

Purpose: Amplify arrived DNA parts.

Procedure:

1. Centrifuge primer tubes for 16000 g 1 minute.
2. Add nucleus free water base on the corresponding data sheet to dissolve the primer.
3. Dilute 10x (3 μ L primer + 27 μ L nucleus free water) each to a new tube.
4. Prepare the following:

Reagent	1 rxn (50 μ L)	24 rxn (1200 μ L)
10X ExTaq buffer	5 μ L	120 μ L
25 μ M dNTP mix	4 μ L	96 μ L
10 μ M forward primer	1 μ L	24 μ L
10 μ M reverse primer	1 μ L	24 μ L
DNA temp	1 μ L	24 μ L
PCR grade water	37.75 μ L	906 μ L
ExTaq polymerase	0.25 μ L	6 μ L

5. These are the following templates:

	Name	Concentration (ng/ μ L)
A	ccaR	119
B	ccaSm3	187
C	Csn: AmilCP	305
D	Csn: ACCD	278
E	Csn: LEA	195
F	Ho1-pcyA	211
G	PcpcG2-veg	90
H	Pveg (STK $\rightarrow$ Moco)	280

6. Distribute 47 μ L of 24 rxn mixture into each tube.
7. Add DNA template and their corresponding primers.
8. Place into PCR machine for the thermal cycle as below:

Thermal cycle number	Temperature (deg C)	Time (sec)
1	98	30
2	98	10
3	53	30
4	72	30
5	72	120
6	25	0

Repeat thermal cycles 2, 3, and 4 for 25 times.

Done by Abby T, Abigail, Abby K, Audrey H, and Sarah.
ELN by Sophia Y.

Date: 20260619

Title: Making 1% agarose gel.

Purpose: Make 24 well agarose gel for gel electrophoresis.

Procedure:

1. Prepare 0.4 g of agarose powder and 40 mL of 1X TAE in a flask.
2. Cover with a piece of plastic wrap and use a pipette tip to poke 3-5 holes.
3. Microwave for 30 sec then agitate it. Repeat this step until there's no particles in the melted gel.
4. Remove the plastic wrap and let the gel cool down until warm but not too hot to touch.
5. Add 2 μ L of safety dye into the gel and agitate it.
6. Pour into the assembled rectangular mold with 24 wells.
7. Use a pipette tip to press out bubbles.
8. Leave to solidify at room temperature.

Done by Abby K, Abby T, and Abigail.
ELN by Sophia Y.

Date: 20260619

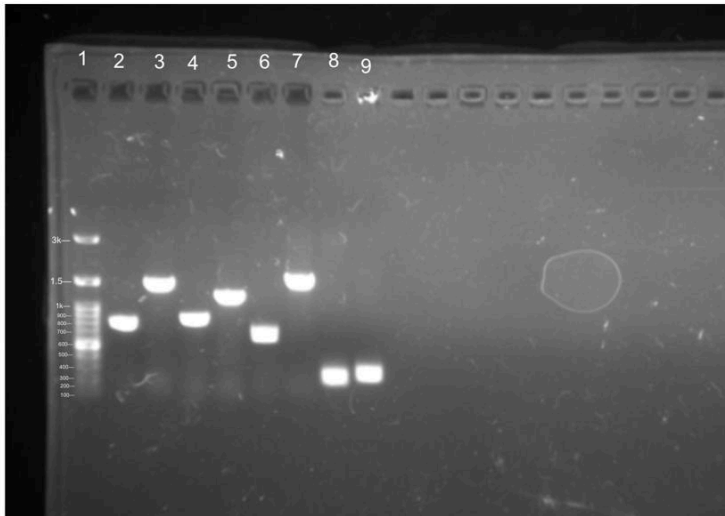
Title: Electrophoresis

Purpose: Verify the size of the PCR product.

Procedure:

1. Mix each of the samples (5 μ L) with 1 μ L of loading dye.
2. Load the samples in the following layout:
M A B C D E F G H
5 5 5 5 5 5 5 5 5
3. Resolve the samples at 100 V for 30 minutes.

20260619 primer PCR



1. [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
2. ccaR
3. ccaS
4. csn:AmilCP
5. csn:AccD
6. csn:Lea
7. Ho1-PcyA
8. PcpG2-veg
9. Pveg (STK->Moco)

Done by Abby T and Abby K.
ELN by Sophia Y.

2026/6/20



Date: 20260620

Title: Cleaning up

Purpose: To isolate target cell from unwanted cell

Procedure:

1. Prepare 8 1.5 mL centrifuge tubes and add in 750 μ L of GB buffer with 150 μ L of PCR product; mix well.
2. Place a 6B column on a collection tube; add 500 μ L of mixture in.
3. Centrifuge at 10,000 g for 30 seconds. Discard flow through.
4. Repeat steps 2 and three.
5. Add 0.5 mL WN Buffer and centrifuge for 30 seconds; discard flow through.
6. Add 0.5 mL WS Buffer and centrifuge for 30 seconds; discard flow through.
7. Centrifuge at full speed for 3 minutes to remove ethanol residue.
8. Place the column on a new 1.5 mL centrifuge tube and add 30 μ L of Elution Buffer.
9. Store the column for 3 minutes and centrifuge at full speed for 2 minutes.
10. Store DNA at 4 deg C or -20 deg C.

Done by Ryan Y and Olivia.

ELN by Sophia Y.

Date: 20260620

Title: 3 liters $\frac{1}{2}$ MS Medium

Purpose: Prepare medium for hydroponic system.

Procedure:

1. Prepare the following:
 - a. $\frac{1}{2}$ MS Medium Powder 3.45 g x 2
 - b. MES 0.75 g x 2
 - c. RO Water 3 L
2. Have 2 2L beakers filled with 1.5 L of water (measuring cylinder).
3. Add in 3.45 g $\frac{1}{2}$ MS medium powder & 0.75 g MES in both beakers; mix well.
4. Tune the pH with KOH until it reaches 5.7-5.8.

	Original pH	Final pH
1st beaker	4.20	5.70
2nd beaker	4.26	5.70

5. Cover the beakers with aluminum foil and autoclave for 20 minutes at 127°C.

Done by Ryan Y.

ELN by Sophia Y.

Date: 20260620

Title: 10 mL $\frac{1}{2}$ MS Agar

Purpose: For hydroponics systems.

Procedure:

1. Prepare:
 - a. Agar powder 0.1 g
 - b. Sucrose 0.1 g
 - c. MES 0.005 g
 - d. ½ MS Medium 0.023 g
 - e. RO water 10 mL
2. Mix all in a beaker and mix well.
3. Tune the pH with KOH until reach 5.7-5.8.
 - a. Original pH: 4.8
 - b. Final pH: 5.71
4. Cover the beaker and autoclave for 20 minutes at 127°C.

Done by Ryan Y.

ELN by Sophia Y.

Date: 20260620

Title: Miniprep

Purpose: Extracting plasmid dna from STK

Procedure:

1. Label the following:

Label	Part Name	Concentration (µg/ml)	A260/280
C1	pSTK-0-sfGFP	442.0	2.167
C2	pSTK-1A-sfGFP	788.0	2.153
C3	pSTK-1B-sfGFP	310.0	2.168
C4	pSTK-1C-sfGFP	305.0	2.133
C5	pSTK-1D-sfGFP	369.0	2.133
E4	pSTK-0B-TM_RBS-2	353.0	2.139
E5	pSTK-0B-TM_RBS4	560.0	2.154
E6	pSTK-0B-PET_RBS	344.0	2.150
E8	pSTK0-B-RBS2	400.0	2.126
E9	pSTK0B-RiboJ-RBS0	355.0	2.126

2. Transfer 1.8 mL of TB medium to a 2 mL microtube
3. Centrifuge for 60 seconds at 16,000 g

4. Discard the supernatant
5. Repeat steps 2-3 until there is no TB left in the tube
6. Add 200 μL of MX1 buffer
7. Resuspend the pellet through vortexing or pipetting
8. The cells should be completely resuspended before moving to step 9.
9. Add 250 μL of MX2 buffer.
10. Gently mix the buffer well by inverting the microtube back and forth for 8-10 times.
11. Incubate the mixture at room temperature for 120 seconds.
12. Add 350 μL of MX3 buffer to the mixture.
13. Immediately and gently mix via inversion 4-5 times.
14. Centrifuge the mixture at max rcf [23,444 g] for 480 seconds.
15. While centrifuging, assemble the column and collection tube.
16. Transfer the supernatant (around 800 μL) to the column using a 200 μL micropipette to avoid transferring/contaminating the supernatant with the pellet.
17. Centrifuge the column at 7,000 g for 30 seconds.
18. Discard the flow-through liquid in the collection tube.
19. Add 500 μL WN buffer.
20. Centrifuge at 7,000 g for 30 seconds.
21. Discard the flow-through.
22. Add 700 μL WS buffer.
23. Centrifuge at 7,000 g for 30 seconds.
24. Discard the flow-through.
25. Centrifuge the empty column at 10,000 g for 3 minutes (next time 8 minutes,) to remove residual ethanol.
26. Place the column into a new 1.5 mL microtube.
27. Add 50 μL sterile distilled water on the center of the membrane.
28. Incubate the column at room temperature for 3 minutes.
29. Centrifuge at 10,000 g for 3 minutes.

Done by Phoebe Chen & Ethan Chang

Supervised by Gabriel Ng

ELN by Ethan Chang

Date: 20260620

Title: Enzyme Digestion

Purpose: Testing enzymatic reaction with newly-miniprep DNA

Procedure:

1. Prepare the mastermix:

Materials	Volume
rCutSmart	20 μL
EcoR1	5 μL

ddH ₂ O	100 µL
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2. Place the contents into a 1.5 mL microtube (EcoR1 last)
3. Mix the contents via pipetting, and distribute 15 µL into 10 separate 0.5 mL microtubes
4. Add 5 µL of DNA to the 0.5 mL microtube [from 20260620 Miniprep]
 - a. Add 5 µL of DNA to the tube and label it respectively to its STK name ('pSTK...')
5. Incubate the labeled 0.5 mL microtubes in the MiniAmp thermocycler at 37C indefinitely until the laboratory is ready to experiment a gel electrophoresis to size the STK DNA

Done by Phoebe and Ethan Chang

Supervised by Gabriel

ELN by Ethan Chang

2026/6/21

Date: 20260621

Title: Miniprep

Purpose: Prepare for electrophoresis

Procedure:

B3: pSTK-1A-5'-lacA

B4: pSTK-1D-3'-lacA

B5: pSTK-1A-5'-sacA

B6: pSTK-1D-3'-sacA

B7: pSTK-1A-5'-xvnP

B8: pSTK-1D-3'-xvnP

B9: pSTK-1A-Spacer

B10: pSTK-1B-Spacer

B11: pSTK-1C-Spacer

B12: pSTK-1D-Spacer

1. Transfer 1.8 mL of TB Medium to 2 mL microtube.
2. Centrifuge for 1 min at 16,000 g.
3. Discard the supernatant.
4. Repeat steps 1-3 until collected all medium.
*Make sure the tip inside the TB medium tube is cleared.
5. Add 200 μ L MX1 buffer.
6. Resuspend the pellet through vortexing or pipetting; cells should be completely resuspended.
7. Add 250 μ L of MX2 buffer.
8. Gently mix by inverting the microtube 8-10 times.
9. Incubate the mixture at room temperature for 2 minutes.
10. Add 350 μ L MX3 buffer.
11. Immediately and gently mix by inverting 4-5 times.
12. Centrifuge at 16,000 g for 5 minutes; meanwhile, place a Mini column onto a collection tube.
13. Transfer the supernatant carefully into the column.
14. Centrifuge at 7,000 g for 30 seconds. Discard the flow-through.
15. Wash the column once with 0.5 mL of WN Buffer by centrifuging at 7,000 g for 30 seconds. Discard the flow-through.
16. Wash the column once with 0.7 mL of WS buffer by centrifuging at 7,000 g for 430 seconds. Discard the flow-through.
17. Centrifuge the column at 10,000 g for another 3 minutes.
18. Place the column into a new 1.5 mL centrifuge tube.
19. Add 50 μ L of elution buffer onto the center of the membrane.
20. Stand the column for 2-3 minutes
21. Centrifuge at 10,000 g for 2-3 minutes to elute DNA.

Results:

	Concentration (ng/ μ L)	A260/A280
B3	134	1.836
B4	134	2.000
B5	139	1.904
B6	160	1.818
B7	160	1.951
B8	154	1.974
B9	148	1.827
B10	168	1.931
B11	178	1.894
B12	160	1.905

Done by Abby T and Abigail.

Supervised by Chars.

ELN by Sophia Y.

Date: 20260625

Title: 1.0% Agarose Gel

Purpose: Prepare 1.0% agarose gel for electrophoresis.

Procedure:

1. Assemble the mold and tray; prepare the comb.
2. Weight 0.2 g of agarose powder and obtain 20 mL 1X TAE solution with a graduated cylinder.
3. Place the contents into a conical flask.
4. Cover the flask with a plastic wrap then use a tip to puncture a few holes in the plastic wrap.
5. Microwave for 30 sec, then take it out and agitate it.
6. Microwave for 10 sec, then take it out and agitate it. Repeat until there are no particles in solution.
7. Wait for the gel to not smoke, then add 1 μ L of safety dye and agitate it.
8. Pour the gel into the tray, then assemble the comb.
9. Wait for the gel to solidify.

Done by Sophia Y.

Recorded by Sophia Y.

Supervised by Chars.

ELN by Sophia Y.

Date: 20260625

Title: Golden Gate Assembly

Purpose: Amplify arrived DNA parts

Procedure:

1. Prepare the reaction as follow

A: (STK) Module 1: Pveg-RBS0-Ho1::pcyA-B0015

Vector 1A	0.176348 μ L
Insert promoter (Pveg)	0.109618 μ L
Insert RBS (RBS0)	0.1461117 μ L
Insert CDS (Ho1::PcyA)	0.09070 μ L
Insert terminator (B0015)	0.268869 μ L
T4 DNA ligase buffer	2 μ L
Sterile water	15.208353 μ L
T4 DNA ligase	1 μ L
Bsal-HFV2	1 μ L
Total	20 μ L

B: (STK) Module 2-1 (PliaG-RBS0-CcaSm3-B0015)

Vector 1B	0.1343046 μ L
Insert promoter (PliaG)	0.2205 μ L
Insert RBS (RBS0)	0.1461117 μ L
Insert CDS (CcaSm3)	0.071598 μ L
Insert terminator (B0015)	0.268869 μ L
T4 DNA ligase buffer	2 μ L
Sterile water	15.1586167 μ L
T4 DNA ligase	1 μ L
Bsal-HFV2	1 μ L

Total	20 μ L
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C: Module(2-2): Plep-RBS0-CcaSm3-B0015

Vector 1B	0.1343046 μ L
Insert promoter (PlepA)	0.18795 μ L
Insert RBS (RBS0)	0.1461117 μ L
Insert CDS (CcaSm3)	0.071598 μ L
Insert terminator (B0015)	0.268869 μ L
T4 DNA ligase buffer	2 μ L
Sterile water	15.1911667 μ L
T4 DNA ligase	1 μ L
Bsal-HFV2	1 μ L
Total	20 μ L

D:Module(3):Pveg-RBS0-CcaR-B0015

Vector 1C	0.1428 μ L
Insert promoter (Pveg)	0.109618 μ L
Insert RBS (RBS0)	0.1461117 μ L
Insert CDS (CcaR)	0.0392 μ L
Insert terminator (B0015)	0.268869 μ L
T4 DNA ligase buffer	2 μ L
Sterile water	15.2934013 μ L
T4 DNA ligase	1 μ L
Bsal-HFV2	1 μ L
Total	20 μ L

E: Module(4): PcpcG2-veg-EEBD-MF001-csn:LEA-B0015

Vector 1D	0.153636 μ L
Insert promoter (PcpcG2-veg)	0.047 μ L

Insert RBS (EEBD-MF001)	0.1461117μL
Insert CDS (csn::LEA)	0.03888μL
Insert terminator (B0015)	0.268869μL
T4 DNA ligase buffer	2 μL
Sterile water	15.3455033 μL
T4 DNA ligase	1 μL
Bsal-HFV2	1 μL
Total	20μL

2. Use the PCR machine to run the program below:

temperature	time	cycle
37°C	1.5min	25-45cycle
16°C	3min	
37°C	5min	1cycle
60°C	10min	1cycle

Done by Alex

2026/6/22



Date: 20260622

Title: Golden Gate Assembly

Purpose: Amplify arrived DNA parts

Procedure:

1. Preparing follow reaction:

- A: (STK)Module 1:Pveg-RBS0-H01::PcyA-B0015

Vector 1A STK002-pSTK-1A-sfGFP	1.8μL(dilute 1/10)
Insert promoter(Pveg)	1.1μL(dilute 1/10)
insert+RBS(RBS0)	1.5μL(dilute 1/10)
insert+CDS(Ho1::PcyA)	1.8μL (dilute 1/20)
insert+terminator(B0015)	1.3μL (dilute(1/5))
T4 DNA ligase buffer	2μL
Sterile water	8.5μL
T4 DNA ligase	1μL
Bsa1-HFV2	1μL

- B:(STK) Module 2-1 PliaG-RBS0-CcaSm3-B0015

Vector 1B (sTK002-pSTK-1B-sfGFP)	1.3μL(dilute 1/10)
Insert promoter(PliaG)	1.1μL(dilute 1/5)
insert+RBS(RBS0)	1.5μL(dilute 1/10)
insert+CDS(CcaSm3)	1.4μL (dilute 1/20)
insert+terminator(B0015)	1.4μL (dilute(1/5))
T4 DNA ligase buffer	2μL
Sterile water	9.3μL
T4 DNA ligase	1μL
Bsa1-HFV2	1μL

- C:(STK)Module 2-2 PlepA-RBS0-CcaSm3-B0015

Vector 1B	1.3μL(dilute 1/10)
Insert promoter(PlepA)	1.9μL(dilute 1/5)

insert+RBS(RBS0)	1.5µL(dilute 1/10)
insert+CDS(CcaSm3)	1.4µL (dilute 1/20)
insert+terminator(B0015)	2.7µL (dilute($\frac{1}{5}$))
T4 DNA ligase buffer	2µL
Sterile water	7.2µL
T4 DNA ligase	1µL
Bsa1-HFV2	1µL

- D:(STK)Module 3-1 Pveg-RBS0-CcaR-B0015

Vector 1C	1.4µL(dilute 1/10)
Insert promoter(Pveg)	1.1µL(dilute 1/5)
insert+RBS(RBS0)	1.5µL(dilute 1/10)
insert+CDS(CcaR)	1.2µL (dilute 1/20)
insert+terminator(B0015)	1.3µL (dilute($\frac{1}{5}$))
T4 DNA ligase buffer	2µL
Sterile water	9.5µL
T4 DNA ligase	1µL
Bsa1-HFV2	1µL

- E: Module 4-1 Pveg-EEBD-MF001-csn::LEA-B0015

Vector 1D (sTK002-pSTK-1B-sfGFP)	2.2µL(dilute 1/10)
Insert promoter(EEBD-MF002)	1.7µL(dilute 1/5)
insert+RBS(csn::LEA)	2.5µL(dilute 1/10)
insert+CDS(CcaSm3)	1.9µL (dilute 1/20)
insert+terminator(B0015)	2.5µL (dilute($\frac{1}{5}$))
T4 DNA ligase buffer	2µL
Sterile water	5.3µL
T4 DNA ligase	1µL

Bsa1-HFV2	1μL
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- F: Module 4-2 Pveg-EEBD-MF001-csn::AmilCP-B0015

Vector 1D	2.2μL(dilute 1/10)
Insert promoter(Pveg)	1.7μL(dilute 1/5)
insert+RBS(EEBD-MF001)	2.5μL(dilute 1/10)
insert+CDS(csn::AmilCP)	1.3μL (dilute 1/20)
insert+terminator(B0015)	2.5μL (dilute(1/5))
T4 DNA ligase buffer	2μL
Sterile water	5.9μL
T4 DNA ligase	1μL
Bsa1-HFV2	1μL

- G:(STK) module 4-1 PcpcG2-veg-EEBD-RBS0-csn::LEA-B0015

Vector 1D	1.5μL(dilute 1/10)
Insert promoter(PcpcG2-veg)	1.4μL(dilute 1/5)
insert+RBS(EEBD-RBS0)	2.1μL(dilute 1/10)
insert+CDS(csn::LEA)	1.6μL (dilute 1/20)
insert+terminator(B0015)	1.3μL (dilute(1/5))
T4 DNA ligase buffer	2μL
Sterile water	8.1μL
T4 DNA ligase	1μL
Bsa1-HFV2	1μL

- H:(STK)module 4-2 (PcpcG2-veg-EEBD-RBS0-csn::ACCD-B0015

Vector 1D	1.5μL(dilute 1/10)
Insert promoter(PcpcG2-veg)	1.4μL(dilute 1/5)
insert+RBS(EEBD-RBS0)	1.0μL(dilute 1/10)
insert+CDS(csn::AccD)	1.2μL (dilute 1/20)

insert+terminator(B0015)	1.3µL (dilute($\frac{1}{5}$))
T4 DNA ligase buffer	2µL
Sterile water	9.4µL
T4 DNA ligase	1µL
Bsa1-HFV2	1µL

- l:(STK) module 4-3 pcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015

Vector 1D	1.5µL(dilute 1/10)
Insert promoter(PcpcG2-veg)	1.4µL(dilute 1/5)
insert+RBS(EEBD-RBS0)	1µL(dilute 1/10)
insert+CDS(csn:AmilCP)	1.4µL (dilute 1/20)
insert+terminator(B0015)	1.3µL (dilute($\frac{1}{5}$))
T4 DNA ligase buffer	2µL
Sterile water	9.2µL
T4 DNA ligase	1µL
Bsa1-HFV2	1µL

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

Done by: Chloe, Abby T, Abigail, Abby K
Supervised by: Gabriel
ELN: Abby T, Sophie L

Date: 20260622

Title: Colony PCR of 06/17 (transformation pTWIST-Amp-High SamyQ-TAT-HA-AmiICP)

Purpose: Amplify arrived DNA parts.

Procedure:

1. Prepare the master mix x5 (5 reactions) 1-5 as follows:

Forward Primer	1
Reverse Primer	1
Q-Amp Master Mix	10
ddH ₂ O	8

FW primer: bio brick prefix

Reverse primer: biobrick suffix

2. Dispense Solution (reaction) to 5 strip PCR tubes
3. Pick up a total of 5 colonies (each colony for each rxn) with the tip (circle the chosen colonies on the Petri dish before picking)
4. Releases the tips with the colonies in the PCR tubes and wait for 5 mins
5. Take out the tips and put tubes in PCR machines thermal cycle for 35 cycles:

95°C	5 min
95°C	30 sec
55°C	30 sec
72°C	1 min
72°C	5 min

Done by: Abigail, Abby K, Abby T

Supervised by: Gabriel

ELN: Sophie L

Date: 20260622

Title: Make 1% Agarose Gel

Purpose: Make 1% agarose gel for electrophoresis

Procedure:

1. Prepare 0.40 g agarose powder and 40 mL 1X TAE.
2. Pour it into a flask.
3. Use plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 seconds and agitate.
6. Microwave for 10 seconds and agitate; repeat this step until there are no sediments left in the solution.
7. Prepare the mold, tray, and comb while waiting for the solution to stop smoking.

8. Add 2 μL of safety dye and agitate.
9. Pour gel into the tray.
10. Wait until the gel solidifies.

Done by: Abigail

ELN: Sophie L

2026/6/23



Date:20260623

Title: Gel Electrophoresis of 20260622 Colony PCR Results

Purpose: check the 20260622 colony PCR results

Procedure:

1. Load samples in the following layout:

M	1	2	3	4	5
5	5	5	5	5	5

M: 3kb DNA ladder

For 1-t please refer to page 140

2. Resolve the samples at 100V for 30 min.

20260623 colony PCR



1. [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
2. pTWIST-Amp-High SamyQ-TAT-HA Amil CP
3. pTWIST-Amp-High SamyQ-TAT-HA Amil CP
4. pTWIST-Amp-High SamyQ-TAT-HA Amil CP
5. pTWIST-Amp-High SamyQ-TAT-HA Amil CP
6. pTWIST-Amp-High SamyQ-TAT-HA Amil CP

Done by: Abby K

Supervised by: Gabriel

ELN: Sophie L

Date:20260623

Title: Miniprep

Purpose: to amplify CCP Amil

Procedure:

1. Transfer 1.8mL of medium to a 2mL microtube
2. Centrifuge at 16000 g for 1 min
3. Discard the supernatant
4. Repeat steps 1-3 until collected all medium
5. Add 200µL of MX1 buffer

6. Resuspend the pellet through vortexing or pipetting
7. The cells should be completely resuspended
8. Add 250µL of MX2 buffer
9. Gently mix the buffer well by inverting 8-10 times
10. Incubate the mixture at room temperature for 2 min
11. Add 350µL MX3 buffer
12. Immediately and gently mix by inverting 4-5 times
13. Centrifuge at 23444 g for 8min
14. While centrifuging, assemble the column and collection tube
15. Transfer the supernatant(800µL) to the column using 200µL micropipette
16. Centrifuge at 7000 g for 30sec
17. Discard the flowthrough
18. Add 500µL UN buffer into the column
19. Centrifuge at 7000 g for 30sec
20. Discard the flowthrough
21. Add 700µL WS buffer
22. Centrifuge at 7000 g for 30sec
23. Discard the flow-through
24. Centrifuge the empty column at 10000 g for 9 min to remove residual ethanol
25. Place the column into a new 1.5mL microtube
26. Add 50µL sterile distilled water in the center of the membrane
27. Incubate the column at room temperature for 3 min
28. Centrifuge at 10000 g for 3 min

	Concentration(ng/µL)	A260/230	A260/180
A	70	0.805	1.795
B	101	0.944	1.870
C	134	0.905	1.861
D	104	0.819	1.825
E	103	0.990	1.907

Done by Chloe, Abby K
Supervised by Gabriel

Date:20260623

Title:Colony PCR

Purpose: Amplify DNA

Procedure:

1. Prepare the master mix x10

	1rxn	10rxn
Forward primer	1	10

(Biobrick prefix)		
Reverse primer (CDS+reverse)	1	10
A-amp master mix	10	100
ddH ₂ O	8	80

Plate A

	2rxn
Forward (biobrick prefix)	2μL
Reverse (pcyA)	2μL
A-amp master mix	20μL
ddH ₂ O	16μL

Plate B

	2rxn
Forward (biobrick prefix)	2μL
Reverse (CcASm3)	2μL
A-amp master mix	20μL
ddH ₂ O	16μL

Plate C

	2rxn
Forward (biobrick prefix)	2μL
Reverse (CcaSm3)	2μL
A-amp master mix	20μL
ddH ₂ O	16μL

Plate G

	2rxn
Forward (biobrick prefix)	2 μ L
Reverse (LEA)	2 μ L
A-amp master mix	20 μ L
ddH ₂ O	16 μ L

Plate H

	1rxn
Forward (biobrick prefix)	1 μ L
Reverse (AccD)	1 μ L
A-amp master mix	10 μ L
ddH ₂ O	8 μ L

Plate I

	1rxn
Forward (biobrick prefix)	1 μ L
Reverse (pcyA)	1 μ L
A-amp master mix	10 μ L
ddH ₂ O	8 μ L

2. Dispense solution (reaction into 10 PCR tubes
3. Pick up a total of 10 colonies with the tip
4. Release the tips with the colonies in the PCR tubes+wait 5 min
5. Place tubes in PCR machines

Done by Abby K, Chloe

Date:20260623

Title:prepare 1.0% agarose gel for electrophoresis

Purpose: prepare gel for electrophoresis**Procedure:**

1. Assemble the mole & tray
2. Weigh 0.2g of agarose powder and add it into a flask
3. Add 20mL of 1XTAE solution into the flask
4. Cover the flask with plastic wrap and punctuate some holes
5. Microwave for ~3-seconds and agitate it. Repeat this step until all particles are dissolved
6. Wait until the gel stops smoking but is still hot
7. Add 1 μ L of safety dye into the gel and agitate it
8. Pour the gel into the tray, then assemble the comb
9. Wait until gel dried

Done by Jaiquelyn

Supervised by Gabriel

Date:20260623

Title:Golden Gate assembly

Purpose: To assemble the light sensor module

Procedure:

1. Prepare the reaction using following protocol

V:vecter

P:powder

R:RBS

C:cds

T:terminater

A:Pveg-RBSO-Ho1::pcyA-B0015(1A) (STK)

V	115 μ L	3120bp
P	280 μ L	3361bp
R	188 μ L	3113bp
C	222 μ L	1549bp
T	107 μ L	2214bp
total		11356bp

B: PliaG-RBS0-CcaSm3-B0015(1B) (STK)

V	151 μ L	3120bp
P	130 μ L	2205bp
R	188 μ L	2113bp

C	266μL	1465bp
T	107μL	2213bp
total		11066bp

C: PlepA-RBS0-CcaSm3-B0015(1B) (STK)

V	151μL	3120bp
P	153μL	2241bp
R	188μL	3113bp
C	266μL	1465bp
T	107μL	2213bp
total		11152bp

D:Pveg-RBS0-CcaR-B0015(1C) (STK)

V	142μL	3120bp
P	280μL	2361bp
R	188μL	2113bp
C	253μL	760bp
T	107μL	2213bp
total		10567bp

E:Pveg-EEBD-MF001-csn::LEA-B0015 (1D) (VoClo2.0)

V	114μL	3815bp
P	183μL	271bp
R	168μL	3224bp
C	195μL	2797bp
T	66μL	2520bp
total		12627bp

F:Pveg-EEBD-MF001-csn::AmilCP-B0015(1D)

V	114μL	3815bp
---	-------	--------

P	183μL	271bp
R	168μL	3224bp
C	305μL	2986bp
T	66μL	2520bp
total		12816bp

G:PcpcG2-veg-EEBD-csn::LEA-B0015(1D) (STK)

V	132μL	3120bp
P	65μL	235bp
R	150μL	2392bp
C	212μL	634bp
T	107μL	2213bp
total		8594bp

H:PcpcG2-veg-EEBD-riboj-RBS0-csn::ACCD-B0015 (1D) (STK)

V	132μL	3120bp
P	65μL	235bp
R	150μL	2392bp
C	252μL	1195bp
T	107μL	2213bp
total		9155bp

I: PcpcG2-veg-EEBD-riboj-RBS0-csn::AmilCP-B0015 (1D) (STK)

V	132μL	3120bp
P	65μL	235bp
R	150μL	2342bp
C	150μL	823bp
T	107μL	2213bp
total		8783bp

J:Pveg-EEBD-riboj-RBS0

V	132µL	3120bp
P	280µL	2362bp
R	150µL	2392bp
C	250µL	823bp
T	107µL	2213bp
total		10904bp

2. For STK using Amp and for MoClo 2.0 using Kana
3. Assemble protocol
(37°C, 1min -> 16°C, 1 min)x30 -> 60°C, 5min
4. Transformation protocol follow 2026/06/02
Done by Gabriel

2026/6/24 

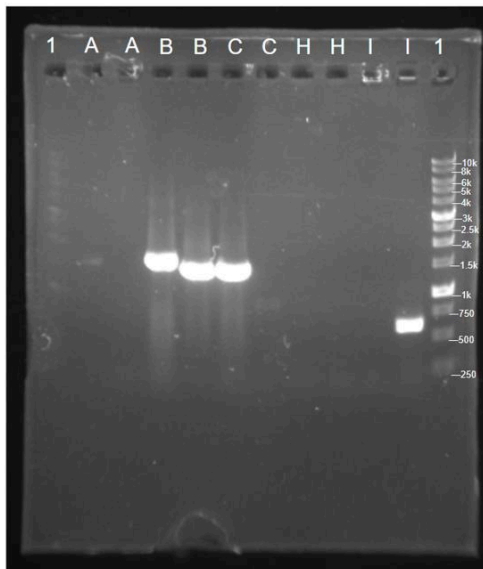
Date: 20260624

Title: Electrophoresis

Purpose: check PCR result

Procedure:

1. Assemble the mold and tray.
2. Fill the tank with 1X TAE buffer to gray line.
3. Load the samples in the following layout:
M A A B B C C H H I I M
5 5 5 5 5 5 5 5 5 5 5 5
1: 1k DNA ladder
For ABC...HI samples, refer to p147~148.
4. Resolve the samples at 100 V for 30 minutes.



1 = Omics 1kb DNA RTU Ladder
A = Pveg - RBS0 - Ho1::PcyA - B0015
B = PliaG - RBS0 - CcaSn3 - B0015
C = PlepA - RBS0 - CcaSn3 - B0015
H = PcpG2veg - EEED - RiboJ - RBS0 - csn::ACCD - B0015
I = PcpG2veg - EEED - RiboJ - RBS0 - csn::AmilCP - B0015

Done by Ethan C.

ELN by Sophia Y.

Date: 20260624

Title: Making 1% agarose gel

Purpose: Confirm colony PCR (page 149) sizes.

Procedure:

1. Collect 0.4 g of agarose powder and 40 mL of 1X TAE.
2. Pour it into a flask.
3. Use a piece of plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 seconds.
6. Take out to agitate.

7. Microwave for 5 seconds.
8. Take out to agitate.
9. Repeat 7-8 until no more sediments.
10. Wait for the mixture to be warm but not smoking.
11. Meanwhile, assemble the tray and comb.
12. Add 2 μL of safety dye.
13. Pour it into the tray.
14. Wait to solidify at room temperature.

Done by Abby K and Sophie L.

Supervised by Chars.

ELN by Sophia Y.

Date: 20260624

Title: TA cloning

Purpose: Insert PCR products (MoClo $\rightarrow$ STK) to pGEM-Teasy.

Procedure:

1. Prepare reactions by the following table:

Code	PCR products	Conc. (ng/ μL)	Insert volume	2x buffer
A	Csn: Lea	21.2	0.6	5
B	Csn: AmilCP	15.0	1.2	5
C	Csn: Accd	25.2	1.1	5
D	CcaSm3	26.6	1.3	5
E	CcaR	25.2	0.7	5
F	Ho1:pcyA	22.2	1.7	5
G	PcpcG2-Pveg	6.5	0.7	5

2. Incubated at 4°C since 23:00.

Done by Chars.

ELN by Sophia Y.

2026/6/25



Date: 20260625

Title: Colony PCR

Purpose: amplify DNA

Procedure:

1. Prepare the master mix (25 reactions).
2. Pipette into 21 μ L of the master mix into 8-strip-PCR-tubes.
3. Circle three colonies from the plates.
A: 1A-Preg-RBS0-Ho1::PcyA-B0015
B: 1C-Preg-RBS0-ccaR-B0015
C: 1D-PcpcG2-veg-EEBD-RBS0-csa::ACCD-B0015
D: 1D-PcpcG2-veg-EEBD-RBS0-csa::LEA-B0015
E: 1B-PlepA-RBS0-ccasm3-B0015
F: 1B-PliaG-RBS0-ccasm3-B0015
G: 1D-PcpcG2-veg-EEBD-RBS0-csa::AmilCP-B0015
H: 1D-Pveg-EEBD-RBS0-csn::AmilCP-B0015
4. Mark the agar plate (ampicillin) by dividing it into 3 rows of 8.
5. For each colony, pick it up with a pipette tip and swipe it on its respective spot on the agar plate, then release the tip into its respective microtube in the 8-strip-PCR-tube.

Done by Sophie L, Sophia Y

Recorded by Sophia Y

Supervised by Venus

Date: 20260625

Title: Making 1% Agarose Gel

Purpose: Make 1% agarose gel to confirm colony PCR sizes using electrophoresis.

Procedure:

1. Prepare 0.40 g agarose powder and 40 mL 1X TAE.
2. Pour it into a flask.
3. Use plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 seconds and agitate.
6. Microwave for 10 seconds and agitate; repeat this step until there are no sediments left in the solution.
7. Prepare the mold, tray, and comb while waiting for the solution to stop smoking.
8. Add 2 μ L of safety dye and agitate.
9. Pour gel into the tray.
10. Wait until the gel solidifies.

Done by Sophie L and Sophia Y.

Recorded by Sophia Y.

ELN by Sophia Y.

Date: 20260625

Title: SOC Medium Preparation**Purpose: prepare SOc medium****Procedure:**

1. Prepare 3.17 g of SOC broth powder.
2. Prepare 100 mL ddH₂O
3. Mix together by using a stirring rod.

Done by Chars.

Recorded by Chloe.

ELN by Sophia Y.

Date: 20260625**Title: Gel electrophoresis****Purpose: Verify page 153 colony PCR success.****Procedure:**

1. Load the gel with the following layout:

M	A1	A2	A3	B1	B2	B3	B4	C1	C2	C3	D1	D2	D3	E1	E2	E3	F1	F2	F3	G1	G2
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
G3	H1	H2	H3																		
5	5	5	5																		

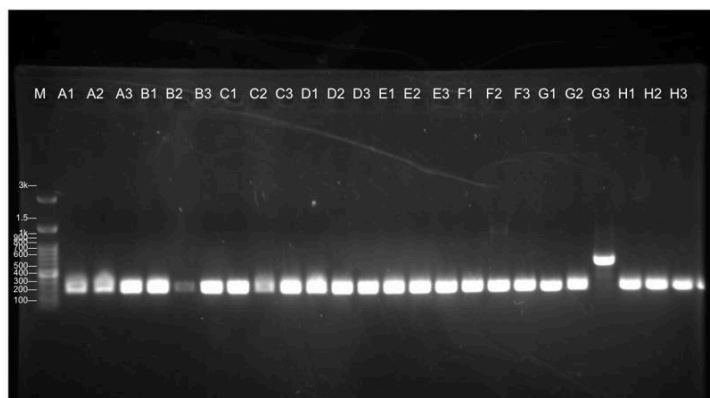
M: 3 kb DNA ladder

A-H: 3 samples of each, refer to page 153

Volume: µL

2. Resolve at 100V for 30 minutes.

20260625 Colony PCR



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
A: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015 (bp)
B: pSTK-1C-Pveg-RBS0-CcaR-B0015 (bp)
C: pSTK-1D-PcpcG2-veg-EEBD-Csn:Accd-B0015 (bp)
D: pSTK-1D-PcpcG2-veg-EEBD-Csn:Lea-B0015 (bp)
E: pSTK-1B-PlepA-RBS0-CcaSm3-B0015 (bp)
F: pSTK-1B-PliaG-RBS0-CcaSm3-B0015 (bp)
G: pSTK-1D-PcpcG2-veg-EEBD-Csn:AmilCP-B0015 (bp)
H: pSTK-1D-Pveg-EEBD-Csn:AmilCP-B0015 (bp)

Done by Abby K.

Supervised by Chars.

ELN by Sophia Y.

Date: 20260625

Title: Transformation

Purpose: Transfer liquid culture to agar plate

Procedure:

1. Turn on the water bath, set it to 42°C and prepare ice
2. Mix 100µL competent cell with 10µL plasmid (A, B, C, D, E, f, G)
3. Incubate on ice for 20min
4. Prepare reaction for blue-white screening
 - A . 1000µL of ddH₂O+4µL of X-Gal (20ng/mL) (final concentration 80mg/mL)
 - B . Spread on LBA plate
5. Heat shock at 42 °C celsius for 45 sec
6. Incubate on ice for 3min
7. Add 1mL of SOC medium and incubate at 37°C celsius for 1 hour
8. Centrifuge at 5000 g for 5min and remove 1ml of the supernatant
9. Resuspend the cell pellet in the remaining supernatant and spread the mixture onto the LBA plate
10. Place the plates into the incubator after labeling

Done by Chloe, Abby K

Supervised by Chars

Recorded by Sophie H.

2026/6/26



Date: 20260626

Title: Electrophoresis Gel Preparation

Purpose: Prepare gel for electrophoresis

Procedure:

1. Prepare and assemble the mold and comb
2. Weight 0.4g of agarose powder and add 40ml of 1X TAE solution into flask
3. Cover the flask with plastic wrap, then use a tip to puncture holes on the plastic wrap
4. Microwave until the gel stops smoking but is still hot
5. Add 2 μ L of safety dye into the gel and agitate it
6. Pour the melted gel into the tray
7. Wait until the gel cools down

Done by Abby K Sophic L

Supervised by Chars

Date: 20260626

Title: Colony PCR

Purpose: amplify DNA

Procedure:

1. Prepare the following reaction (master mix) X100 (100 reactions) and dispense solution into PCR tubes.

	1 reaction	100 reactions
Forward Primer (pSTKseq-fw 10 mM)	1 μ L	100 μ L
Reverse Primer (pSTKseq-rv 10 mM)	1 μ L	100 μ L
Qamp Master Mix	10 μ L	1000 μ L
ddH ₂ O	8 μ L	800 μ L

2. Circle 12 colonies form each of the plates.
A: 1A-Pveg-RBS0-Ho1::PcyA-B0015
B: 1C-Pveg-RBS0-CcaR-B0015
C: 1D-PcpcG2veg-EEBD-RBS0-csn::ACCD-B0015
D: 1D-PcpcG2veg-EEBD-RBS0-csn:LEA-B0015
E: 1B-PlepA-RBS0-Ccasm3-B0015
F: 1B-PliaG-RBS0-Ccasm3-B0015
G: 1D-PcpcG2veg-EEBD-RBS0-csn:AmilCP-B0015
H: 1D-Pveg-EEBD-RBS0-csn:AmilCP-B0015
3. Pick up 12 colonies per plate with a tip and streak the labelled grid.
4. Dispense the tips with colonies into the PCr strip and wait 5 min.

5. Take out the tips and close the lid.
6. Place into PCR machine for the thermal cycle.

Done by Abigail, Abby T, Abby K, Chloe, Sophie L.

Supervised by Chars.

ELN by Sophia Y.

Date: 20260626

Title: Making 1% Agarose Gel

Purpose: prepare agarose gel for electrophoresis

Procedure:

1. Prepare a 0.20g agarose powder and 20 mL 1X TAE.
2. Pour it into a flask.
3. Use plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 sec and agitate.
6. Microwave for 10 sec and agitate until there are no sediments left.
7. Prepare the mold, tray, and comb while waiting for the solution to stop smoking.
8. Add 1 μ L of safety dye and agitate.
9. Pour into the tray.
10. Wait until the gel solidifies.

Done by Sophie L.

Supervised by Gabriel.

ELN by Sophia Y.

Date: 20260626

Title: Minprep

Purpose: to isolate plasmid

Procedure:

- A: pSTK-1B PliaG: Ccasm3 #1
- B: pSTK-1B PliaG: Ccasm3 #2
- C: pSTK-1B Plepa: Ccasm3 #1
- D: pSTK-1A Pveg:Ho1:pcyA #3
- E: pSTK-1D PcpcG2-veg:csn:AmilCP #2
- F: pSTK-1C Pveg:csn:AmilCP #1

1. Transfer 1.8 mL of TB medium to 2 mL microtube.
2. Centrifuge for 1 min at 16,000 g.
3. Discard the supernatant.
4. Repeat steps 1-3 until collected all medium.
5. Add 200 μ L of MX1 Buffer to the pellet and resuspend the cells completely by vortexing or pipetting.

6. Add 250 μL of MX2 Buffer.
7. Gently mix by inverting the tube 8-10 times.
8. Incubate at room temp for 2 minutes.
9. Add 350 μL of MX3 Buffer and immediately and gently mix the solution well.
10. Centrifuge at 10,000 g for 5 minutes. Meanwhile, place a MiniPlus column onto a collection tube.
11. Transfer the supernatant onto the column.
12. Centrifuge at 7,000 g for 30 seconds; discard the flow through.
13. Wash the column once with 500 μL of WN buffer by centrifuging at 7,000 g for 30 seconds; discard the flow through.
14. Wash the column once with 700 μL of WS buffer by centrifuging at 7,000 g for 30 seconds. Discard flow through.
15. Centrifuge column at 1,000 g for another 3 mins.
16. Place column into new 1.5 mL centrifuge tube.
17. Add 50 μL of Elution Buffer onto the center of the membrane.
18. Stand the column for 2-3 mins and centrifuge at 10,000 g for 3 mins.
19. Test concentration at Biodrop.

Results:

	Concentration (ng/ μL)	A260/A280
A	78.00	2.053
B	86.00	1.870
C	80.00	2.000
D	98.00	1.922
E	111.0	1.790
F	64.00	1.882

Done by Abigail, Abby T, and Chloe.

Supervised by Chars.

ELN by Sophia Y.

Date: 20260626

Title: Gel electrophoresis

Purpose: check PCR result

Procedure:

1. Load the gel with the following layout:

Gel 1

M	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
---	----	----	----	----	----	----	----	----	----	-----	-----	-----

5	5	5	5	5	5	5	5	5	5	5	5	5
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11		
5	5	5	5	5	5	5	5	5	5	5		

Gel 2

M	B12	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
5	5	5	5	5	5	5	5	5	5	5	5
C11	C12	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
5	5	5	5	5	5	5	5	5	5	5	5

Gel 3

M	D11	D12	E1	E2	E3	E4	E5	E6	E7	E8	E9
5	5	5	5	5	5	5	5	5	5	5	5
E10	E11	E12	F1	F2	F3	F4	F5	F6	F7	F8	F9
5	5	5	5	5	5	5	5	5	5	5	5

Gel 4

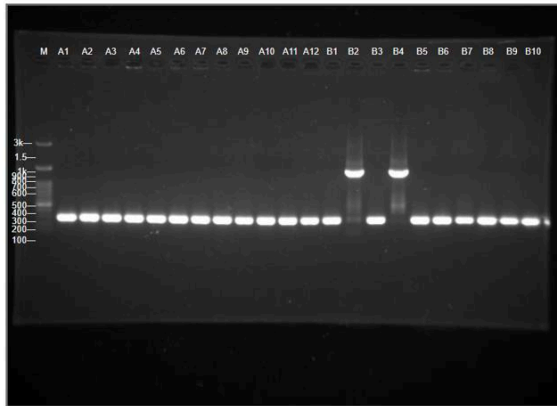
M	F10	F11	F12	G1	G2	G3	G4	G5	G6	G7	G8
5	5	5	5	5	5	5	5	5	5	5	5
G9	G10	G11	G12	H1	H2	H3	H4	H5	H6	H7	H8
5	5	5	5	5	5	5	5	5	5	5	5

Gel 5

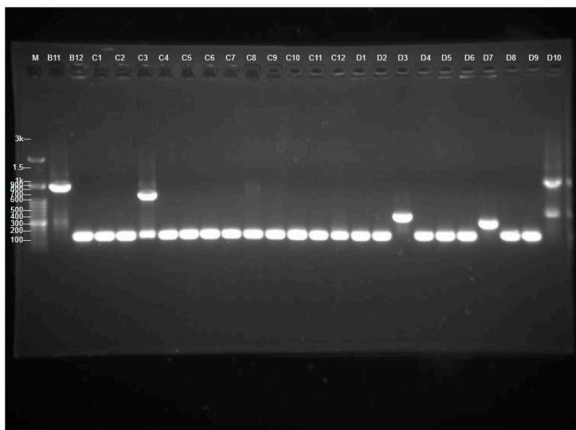
M	H9	H10	H11	H12
5	5	5	5	5

2. Resolve at 100V for 30 minutes.

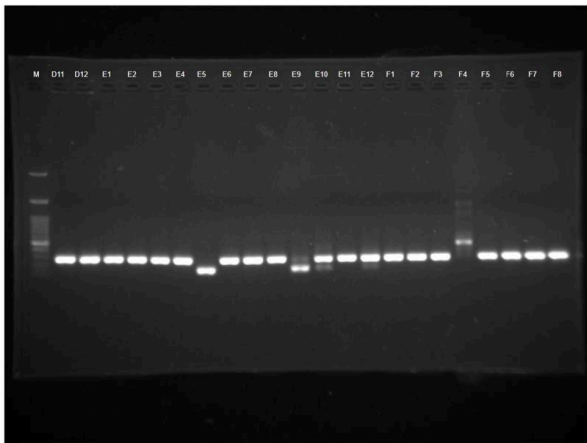
Results:



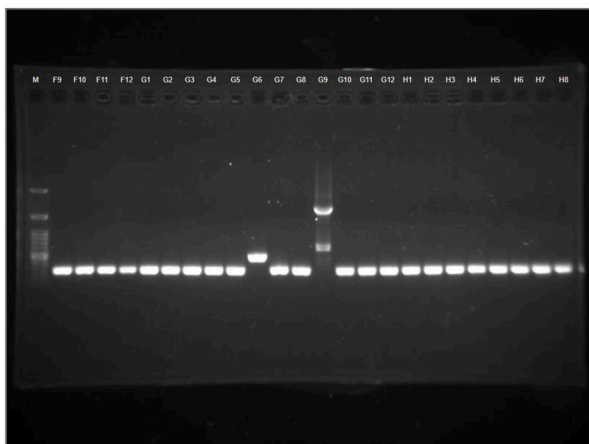
M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
A:1A-Pveg-RBS0-H01::PcyA-B0015 (bp)
B:1C-Pveg-RBS0-CCaR-B0015 (bp)



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
B:1C-Pveg-RBS0-CCaR-B0015 (bp)
C:1D-PcpcG2-veg-EEBD-RBS0-Csn::ACCD-B0015 (bp)
D:1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015 (bp)



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
D:1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015 (bp)
E:1B-PlepA-RBS0-Ccasm3-B0015 (bp)
F:1B-PliaG-RBS0-Ccasm3-B0015 (bp)



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5
uL
F: 1B-PliaG-RBS0-Ccasm3-B0015 (bp)
G: 1D-PcpcG2veg-EEBD-RBS0-Csn:AmilCP-B0015
(bp)
H: 1D-Pveg-EEBD-RBS0-Csn:AmilCP-B0015 (bp)



M: [DM2300] ExcelBand™ 100 bp+3K DNA
Ladder 5 uL
H: 1D-Pveg-EEBD-RBS0-Csn:AmilCP-B0015
(bp)

Done by Sophie L and Sarah C.
ELN by Sophia Y.

2026/6/27

Date: 20260627

Title: Miniprep

Purpose: To isolate DNA for amplification

Procedure:

1. Transfer 1.8mL TB medium to a 2 mL microtube
B2 : 1C-Pveg-RBS0-CCaR-B0015 #2
B4 : 1C-Pveg-RBS0-CCaR-B0015 #4
C3 : 1D-PcpcG2veg-EEBD-RBS0-csn::ACCD-B0015 #3
D10 : 1D-PcpcG2veg-EEBD-RBS0-csn::LEA-B0015 #10
G9 : 1D-PcpcG2veg-EEBD-RBS0-csn::AmilCP-B0015 #9
2. Centrifuge at 16000 g for 1 minute
3. Discard the supernatant
4. Repeat steps 2-3 until there is no TB medium left in the tube
5. Add 200µL of MX1 buffer
6. Resuspend the pellet through vortexing or pipetting
7. The cell should be completely resuspended before moving to the next step
8. Add 250µL of MX2 buffer
9. Gently mix the buffer well by inverting the microtube back and forth 8 times
10. Incube the mixture at room temperature for 2 minutes
11. Add 350µL of MX3 buffer to the mixture
12. Immediately and gently mix via inversion 4-5 times
13. Centrifuge the mixture at max ref(23.44 g) for 5 minutes
14. While centrifuge , assemble column and collection
15. Transfer supernatant 1800 µL to the column using a 200 µL micropipette to avoid transferring pellet and contaminating the supernatant
16. Centrifuge the column at 7000g for 30 min
17. Discard the flow-through liquid in the collection tube
18. Add 500 µL of WN buffer
19. Centrifuge at 7000g for 30 sec
20. Discard the flow through
21. Add 700 µl WS buffer
22. Centrifuge at 7,000g for 30 seconds
23. Discard the flow-through
24. Centrifuge the empty column at 10,000 g for 3 minutes to remove residual ethanol
25. Place the column into a new 1.5 ml microtubule
26. Add 50 µl of EB buffer on the center of the membrane
27. Incubate the column at room temperature for 3 minutes
28. Centrifuge at 10,000 g for 3 minutes.

Done by Sarah , Ethan Chang , Naomi

Supervised by Chars

Results:

	Concentration (ng/ μ L)	A260/280	A260/230
B2	120.0	1.846	0.930
B4	69.0	1.971	1.353
D10	129.0	1.743	0.772
C3	90.0	1.837	1.011
G9	106.0	1.738	0.768

Date: 20260627**Title: PCR****Purpose: amplify the arrived DNA parts****Procedure:**

1. DNA templates

Code	Name	Concentration (ng/ μ L)	Forward Primer	Reverse primer
A	ccaR	119	ccaRF	ccaR R
B	ccaSm3	187	Ccaw3	ccaSm3 R
C	csn:AmilCP	305	csnF	AmilCP R
D	csn::ACCD	278	csnF	ACCD R
E	csn::LEA	195	csnF	LEA R
F	Ho1-pcyA	211	H01F	Pcy A R
G	PcpcG2-veg	90	PcpcG2-vegF	PcpcG2-veg R
H	Cpp Amil 1	70	SamyQ	AmilCP R

2. Prepare the following

reagent	1rxn(50 μ L)	3rxn(150 μ L)x9 (master mix)
10x ExTaq buffer	5 μ L	15 μ L
2.5pM dNTP mix	4 μ L	12 μ L
10 μ M forward primer	1 μ L	3 μ L

DNA temp	1μL	3μL
10μM reverse primer	1μL	3μL
PCR grade water	37.75μL	113.25μL
Ex Taw polymerase	0.25μL	0.75μL

3. Distribute 50μL of 3rxn mixture into each tube for the corresponding part
4. Place into PCR machine, applying the thermalcycle below:

98°C	30sec
98°C	10sec
60°C	30sec
72°C	90sec
72°C	5min
23°C	0min

Done by Audrey H, Abby T

Supervised by Chars

Date: 20260627

Title: Prepare 1% agarose gel

Purpose: extracting desired DNA gel electrophoresis

Procedure:

1. Assemble the mold and comb
2. Weight 0.2g of agarose powder and 20mL of 1XTAE into a flask
3. Cover the flask with a plastic wrap and poke 3 holes with a tip
4. Put the flask into the microwave for 30 sec
5. Take it out and agitate it
6. Repeat step 3 and 4 until all particles are dissolved
7. Take the flask out and cool down until its warm but still hot
8. Put 1 μL of safety dye into the flask
9. Pour the melted gel into the mold
10. Wait 20 min for the gel to solidify
11. Repeat steps 1-10 until 8 preces of gel are made

Done by Audrey H, Abby T, Abby K , Sophie L

Supervised by : Chars

Date: 20260627

Title: Gel electrophoresis

Purpose: check the PCR result

Procedure:

1. Mix 50µL of PCR product from 1 tube with 10µL of loading dye
2. Load samples in the following layout

Gel1

M A M

5 150 5

Gel2

M B M

5 150 5

Gel3

M C M

5 150 5

Gel4

M D M

5 150 5

Gel5

M E M

5 150 5

Gel6

M F M

5 150 5

Gel7

M G M

5 150 5

Gel8

M H M

5 150 5

1:1kb DNA ladder

A:CcaR

B:CcaSm3

C:can:AmilCP

D:can:AccD

E:can:LEA

F:H01:pcya

G:PcpcG2-veg

h:CcpAml1

3. Resolve at 100v for 30,in

Done by Abby K/T, Sophie L

Supervised by Chars

Date: 20260627

Title: gel extraction

Purpose: isolate desired gene fragments from PCR product

Procedure:

1. Under the UV light, identify if the DNA is the correct one
2. Use a knife to cut out the DNA bind
3. Cut up the vine into smaller pieces
4. Place the pieces into the 2mL microtube
5. Add 500 μ L of GP buffer
6. Place the tube onto the thermal shaker at 800rpm, 60°C

Done by Abby K/T, Sophie L

Supervised by Gabriel, Chars

2026/6/28



Date: 20260628

Title: Miniprep

Purpose: isolate the plasmid

Procedure:

G4: pSTK-OC2-His6x

H7 : pSTK-OCx-Spacer

E8 : pSTK0-B-RBS2

H10 : (cPCR)

1. Transfer 1.8mL of TB medium to 2mL microtub
2. Centrifuge for 1 min at 16000 g
3. Discard the supernatant
4. Repeat steps 1-3 until collected all medium
5. Add 200 μ L of MX1 buffer
6. Resuspend the pellet through vortexing or pipetting
7. Add 250 μ L of MX2 buffer
8. Gently mix by inverting the microtube 8-10 times
9. Incubate the mixture at room temperature of room temperature for 2 minutes
10. Add 350 μ L MX3 buffer
11. Immediately and gently mix through inverting 4-5 times
12. Centrifuge at 16000 g for 5 minutes
13. Meanwhile, assemble the column and collection tube
14. Transfer the supernatant (around 800 μ L) to the column using 200mL pipette
15. Centrifuge at 7000 g for 30 second, discard the flow through
16. Wash the column once with 500 μ L of WN buffer by centrifuging at 7000 g for 30 seconds , discard the flow through
17. Centrifuge column into new 1.5mL centrifuge tube
18. Add 50 μ L of nuclear free water onto the center of membrane
19. Stand the column for 2-3 ins and centrifuge at 10000 g for 3 min
20. Test concentration at biodrop

	Concentration(ng/ μ L)	A260/230	A260/280
G4	135.0	1.216	1.957
H7	127.0	0.722	1.868
E8	78.00	1.200	1.950
H10(cpCR)	98.06	0.907	1.885

Done by Abigail

Supervised by Chars

Date: 20260628

Title: Making liquid culture

Purpose: Amplify F2 and F8 from the subtiTookit

Procedure:

1. Label two liquid culture tubes with F2 and F8
2. Turn on the bunsen burner
3. Add 3 mL of LB medium into each tube
4. Add 5 μ L of Chloramphenicol into each tube
5. Dip a medium pipette tip into the subtiToolKit respective agar
6. Dispense the tips(with agar) in their respective tubes
7. Place the tubes into the incubator at a tilted angle

Done by Abby K

Supervised by Gabriel , Chars

2026/6/29

Date: 20260629

Title: Golden Gate Assembly

Purpose: To assemble multiple core genetic parts of the light-controlled optogenetic circuit from level 1 and level 2.

Procedure:

1. Insert: vector=2:1 pmols

A: (STK) Module 1(1A): Pveg–RBS0-Ho1:pcyA-B0015

1. prepare the reaction as follow:

Vector (1A)	1.8 µL (dilute 1/10)
Insert promoter (Pveg)	1.1 µL (dilute 1/10)
Insert RBS (RBS0)	1.5 µL (dilute 1/10)
Insert CDS (Ho1::PcyA)	1.7 µL (dilute 1/20)
Insert Terminator (B0015)	1.3 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL
Sterile water	8.6 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total	20 µL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
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37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

B: M4-Pveg - EEBD- RBS0 - CSN::ACCD - B0015

1. prepare the reaction as follow:

Vector (1D)	1.5 µL(dilute 1/10)
Insert promoter (Pveg)	1.1 µL (Dilute 1/10)
Insert RBS (EEBD-RBS0)	2.1 µL(dilute 1/10)
Insert CDS (CSN::ACCD)	1.5 µL(dilute 1/10)
Insert Terminator (B0015)	1.3 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL
Sterile water	8.5 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total	20 µL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

C: M3-Pveg-RBS0-CcaR-B0015

1. prepare the reaction as follow:

Vector (1C)	1.4 µL(dilute 1/10)
Insert promoter(Pveg)	1.1 µL (dilute 1/10)
Insert RBS(RBS0)	1.5 µL (dilute 1/10)
Insert CDS(CcaR)	1.0 µL (dilute 1/10)
Insert Terminator (B0015)	1.3 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL
Sterile water	9.6 µL
T4 DNA ligase	1 µL

Bsal-HFv2	1 μ L
Total	20 μ L

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of Bsal restriction enzyme)	10 minutes	1 cycle

D: Pveg-EEBD-RBS0-SamyQ::TAT-Ha::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-1-sfGFP (bp:6929)	1.4 μ L (dilute $\frac{1}{5}$)
Insert promoter(Pveg)	1.1 μ L (dilute 1/10)
Insert RBS(EEBD-RBS0)	1.0 μ L (Dilute 1/5)
Insert CDS(SamyQ::TAT-Ha::AmilCP)	1.1 μ L (dilute $\frac{1}{5}$)
Insert Terminator (B0015)	1.3 μ L (dilute 1/5)
T4 DNA ligase buffer	2 μ L
Sterile water	10.0 μ L

T4 DNA ligase	1 μ L
BsaI-HFv2	1 μ L
Total	20 μ L

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

E: Spacer-Spacer-Spacer-Pveg-EEBD-RBS0-Csn::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-2-sfGFP	1.6 (dilute $\frac{1}{5}$)
Insert transcription unit 1(Spacer A): pSTK-1A-Spacer	1.9 μ L (dilute 1/10)
Insert transcription unit 2(Spacer B): pSTK-1B-Spacer	1.7 μ L (dilute 1/10)
Insert transcription unit 3(Spacer C): pSTK-1C-Spacer	1.6 μ L (dilute 1/10)
Insert transcription unit 4(Pveg-EEBD-RBS0-Csn::AmilCP-B0015)	2.4 μ L (dilute $\frac{1}{5}$)
T4 DNA ligase buffer	2 μ L

Sterile water	6.9 μL
T4 DNA ligase	1 μL
Bsmb1	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

F: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-1-sfGFP - concentration($\text{ng}/\mu\text{L}$): 162 - bp: 6929	1.4 μL (dilute $\frac{1}{5}$)
Insert promoter(Pveg) - concentration($\text{ng}/\mu\text{L}$): 280 - bp: 2361	1.1 μL (dilute 1/10)
Insert RBS(EEBD-RBS0) - concentration($\text{ng}/\mu\text{L}$): 188 - bp: 2113	2.1 μL (dilute 1/10)

Insert CDS(GanA-Gala) • concentration(ng/μL): 177 • bp: 2080	1.5 μL (dilute 1/10)
Insert Terminator (B0015) • concentration(ng/μL): 107 • bp: 2213	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	8.6 μL
T4 DNA ligase	1 μL
Bsal-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of Bsal restriction enzyme)	10 minutes	1 cycle

2. Pmol(0.05) insert: vector=1:1

D: Pveg-EEBD-RBS0-SamyQ::TAT-Ha::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-1-sfGFP (bp:6929)	1.4 μ L
Insert promoter(Pveg)	1.1 (dilute 1/4) (1 μ L insert in 3 μ L water)
Insert RBS(EEBD-RBS0)	1.1 (dilute 1/3) (1 μ L insert in 2 μ L water)
Insert CDS(SamyQ::TAT-Ha::AmilCP)	1.1 (dilute 1/2) (1 μ L insert in 1 μ L water)
Insert Terminator (B0015)	1.3 (dilute 1/2) (1 μ L insert in 1 μ L water)
T4 DNA ligase buffer	2 μ L
Sterile water	9.94981071 μ L
T4 DNA ligase	1 μ L
bsa1	1 μ L
Total	20 μ L

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

E: Spacer-Spacer-Spacer-Pveg-EEBD-RBS0-Csn::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-2-sfGFP	1.6 μ L
Insert transcription unit 1(Spacer A): pSTK-1A-Spacer	1.4 μ L(dilute 1/3) (1 μ L insert in 2 μ L water)
Insert transcription unit 2(Spacer B): pSTK-1B-Spacer	1.2 μ L(dilute 1/3) (1 μ L insert in 2 μ L water)
Insert transcription unit 3(Spacer C): pSTK-1C-Spacer	1.2 μ L(dilute 1/3) (1 μ L insert in 2 μ L water)
Insert transcription unit 4(Pveg-EEBD-RBS0-Csn::AmilCP-B0015)	1.2 μ L
T4 DNA ligase buffer	2 μ L
Sterile water	9.4 μ L
T4 DNA ligase	1 μ L
bsmb1	1 μ L
Total	20 μ L

3. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	

37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

F: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. Prepare the reaction as follows:

Vector (shuttle vector): pSTK-EXP-1-sfGFP • concentration(ng/μL): 162 • bp: 6929	1.4 μL
Insert promoter(Pveg) • concentration(ng/μL): 280 • bp: 2361	1.1 (dilute 1/4) (1 μL insert in 3 μL water)
Insert RBS(EEBD-RBS0) • concentration(ng/μL): 188 • bp: 2113	1.1 (dilute 1/3) (1 μL insert in 2 μL water)
Insert CDS(GanA-Gala) • concentration(ng/μL): 177 • bp: 2080	1.1 (dilute 1/3) (1μL insert in 2 μL water)
Insert Terminator (B0015) • concentration(ng/μL): 107 • bp: 2213	1.3 (dilute 1/2) (1 μL insert in 1 μL water)
T4 DNA ligase buffer	2 μL
Sterile water	10.0 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

Done by Abigail, Abby T, and Chloe

Date: 20260629

Title: 500mL agar plate (hydroponics)

Purpose: prepare arabidopsis for hydroponic testing

Procedure:

1. Prepare the following

½ MS medium powder (with sucrose)	1.15g
MES buffer	0.25g
ddH ₂ O	500mL
sucrose(1%)	5g
Agar powder(1%)-ad last	5g

2. Add contents except agar powder into 500mL flask
3. Stir with hotplate & magnet until clear of sediment
4. Use plt probe to measure pH of sodium (pH5.7~5.8)
 - Add 0.1M KOH if pH level is low
5. Distribute to 2 500mL flasks
6. Add 2.5g agar powder each
7. Stir until no sediments
8. Wrap necks with aluminum, tape autoclave indicator
9. Autoclave for 20 minutes at 127°C
10. Transfer flasks to BSC
 - Store in hot bath if not immediately pared

11. Dry clean 13sec surface with wipes 10% FtH
 12. 50mL agar per plate
- Done by Ethan C
Supervised by Gabriel

Date: 20260629

Title: Miniprep

Purpose: isolate DNA for exemplification (F2&F8)

Procedure:

1. Transfer 1.8mL TB medium to a 2mL microtube
2. Centrifuge at 16000 g for 1 min
3. Discard the supernatant
4. Repeat steps 1-3 until no TB buffer
5. Add 200µL of MX1 buffer
6. Resuspend pellet via pipetting/vortexing
7. Cells should be completely resuspended before step5
8. Add 250µL of MX2 buffer
9. Gently mix via inversion 8-10 times
10. Incubate at room temperature for 2 min
11. Add 350µL MX3 buffer
12. Immediately invert 4-5 times
13. Centrifuge at max rcf(23444 g) for 5 min
14. Assemble column & collection tube meanwhile
15. Transfer supernatant(800µL) with 200µL micropipette
16. Centrifuge 7kg for 30sec
17. Discard flow-through
18. Add 500µL of WN buffer
19. Centrifuge 7000 g for 30sec
20. Discard flow-through
21. Add 500µL of WS buffer
22. Centrifuge 7000 g for 30sec
23. Discard flow-through
24. Centrifuge empty column 10000 g for 3 min
25. Place column to 1.5mL microtube
26. Add 50µL of EB buffer directly on the membrane
27. Incubate column at room temperature for 3min
28. Centrifuge 10000 g for 3 min

Done by EthanC
Supervised by Gabriel

2026/6/30 

Date: 20260630

Title: LB Agar plate

Purpose: prepare LB agar plate

Procedure:

1. Measure 7.5 g of LB and 4.44g of agar in 300 ml of distilled water.
2. Use a hot plate and stirring rod to stir until no sediments remain in the solution.
3. Autoclave at 127°C for 20 mins.
4. Bring the solution to the BSC and add 300 µl of Chloramphenicol into the solution
5. Pipette the solution 20 ml into each of the petri dishes and leave to dry.

Done by Ryan Wei and Naomi

Recorded by Ryan Wei.

Date: 20260630

Title: Golden Gate Assembly

Purpose: To assemble Pveg-EEBD-RBS0-spacer-GanA-Gala-histag-B0015 (pmol = 0.05, vector:insert = 1:1)

Procedure:

1. Prepare the reaction as follows:
- 2.

Vector pSTK-EXP-1	1.39 µL
Insert Pveg	1.10 µL (dilute 1/4)
Insert EEBD-RBS0	1.10 µL (dilute 1/3)
Insert spacer (STK244-pSTK-0Cx)	1.07 µL (dilute 1/2)
Insert GanA-Gala	1.15 µL (dilute 1/3)
Insert 6x His Tag	1.01 µL (dilute 1/2)
Insert B0015	1.34 µL (dilute 1/2)
T4 DNA ligase buffer	2 µL
Sterile water	7.84 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL

3. Use the thermocycler to run the program below

Temperature	Time (min)	cycles
37°C (restriction enzyme digestion)	1.5	25-45

16°C (ligation)	3	25-45
37°C (restriction enzyme digestion)	5	1
60°C (heat inactivation of Bsal)	10	1

Done by Naomi, recorded by Naomi.
Supervised by Gabriel

Date: 20260630

Title: Transformation

Purpose: pick up single column for sequencing

Procedure:

1. Turn on the water bath, set it to 42°C and prepare ice.
2. Mix 60 µl competent cells with 10 µl plasmid in each of 6 respective tubes.
3. Incubate on ice for 20 min.
4. Heat shock at 42°C for 45 sec.
5. Incubate on ice for 3 min.
6. Add 1 ml of SOC medium and incubate at 37°C for 1 hr.
7. Centrifuge at 5000 g for 5 min and remove 1 ml of the supernatant.
8. Resuspend the remaining mixture and spread the mixture onto the LBA plate.
9. Place the plates into the incubator after labelling.

Done by Ryan Wei, recorded by Ryan Wei.

Date: 20260630

Title: Liquid culture

Purpose: grow and amplify bacterial cells from single colonies

Procedure:

1. Label liquid culture tubes.
2. Turn on the bunsen burner.
3. Add 3 ml of LB medium and 3 µl of chlorphenicol into each tube.
4. Dip a pipette tip into the colonies.
5. Dispense the tip into their respective tubes.
6. Place the tubes into the incubator at a slanted angle.

Done by Ryan Wei, recorded by Ryan Wei.

Date: 20260630

Title: 0.8% Agarose Gel Preparation for Gel electrophoresis

Purpose: Make 13 well agarose gel for gel electrophoresis

Procedure:

1. Prepare and assemble the mold and the comb.
2. Prepare 0.32 g of agarose powder and add it into a flask.
3. Add 40 mL of 1X TAE solution into the flask.
4. Cover the flask with plastic wrap, then use a tip to puncture 3 to 5 holes on the plastic wrap.
5. Microwave for 30 sec then agitate it. Repeat this step until there's no particles in the melted gel.
6. Wait until the gel stops smoking but is still hot.
7. Add 1 μL of safety dye into the gel and agitate it.
8. Pour the melted gel into the tray.
9. Use a pipette tip to press out bubbles.
10. Leave to solidify at room temperature.

Done by Naomi, recorded by Naomi

Supervised by Gabriel